INDIA DATA CENTRE ATLAS · COMPANION DOCUMENT

Formula and Calculation Handbook

Every equation in the atlas, why it takes the form it does, the rationale for every assumed value, and each calculation worked through with the values set on the sliders.

36 calculations across 17 chapters · Calculated from the atlas's default slider settings on 11 October 2026 · Units: MU = million kWh (GWh); ₹ cr = ₹ crore; ML = million litres; MLD = ML per day

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How to read this handbook

Each section follows the same order: the inputs taken from the atlas sliders and buttons; the formulas; what each equation does and why it is built that way; the assumed values with the reason each was chosen and where it comes from; and the calculation worked through step by step with those inputs. Change a slider in the atlas and the same steps produce the new answer; each formula drawer (ƒ how) shows the live version.

BASIS LABELS

Reported published by a named source · Estimate engineering or third-party estimate · Series assumption a modelling choice made in this research series · Physics a physical constant or law · Convention a unit conversion or standard definition.

CONVERSIONS AND CONSTANTS USED THROUGHOUT

TERM	MEANING	WHY	BASIS
`8.76`	MU per MW-year	One MW running all 8,760 hours of a year delivers 8,760 MWh = 8.76 GWh. Indian utilities call a GWh a "MU" (million units of kWh).	Convention
`× 0.1`	₹ crore per (MU × ₹/kWh)	1 MU × ₹1/kWh = ₹10,00,000 = ₹0.1 crore. Used whenever energy (MU) is multiplied by a tariff.	Convention
`× 10`	₹/kWh per (₹ crore ÷ MU)	₹1 crore ÷ 1 MU = $₹10^7 \div 10^6$ kWh = ₹10/kWh. Used to turn yearly cost per MU into a unit cost.	Convention
`MU × L/kWh = ML`	million litres	10^6 kWh × 1 L/kWh = 10^6 L = 1 ML. Lets water intensities be multiplied directly by energy in MU.	Convention
`0.71 t/MWh`	grid CO_2 factor	CEA CO_2 Baseline Database v21 (FY2024-25) weighted average for India's grid. Same value is used across chapters so results are comparable.	Reported
`CRF(r, n)`	$r(1+r)^n \div ((1+r)^n - 1)$	Capital recovery factor: the equal yearly payment that repays ₹1 of capex at interest r over n years (the maths of a loan EMI). It converts one-off capex into a yearly cost that can be added to yearly opex.	Convention

Results at a glance

The final step of each worked calculation, at the default settings.

#	CALCULATION	CHAPTER	FINAL STEP	RESULT
1	Power flow schematic	Overview	Overhead cost	₹53.0 crore/yr
2	Simulator model	▶ Build a data centre	CO_2	1,24,516 t/yr
3	Flow diagrams	▶ Build a data centre	Water make-up	220.8 ML
4	Tier downtime	01 · Types of data centre	Red squares	1
5	Type finder	01 · Types of data centre	rule-based	–
6	Rack inspector	02 · AI vs conventional	Cooling class	Air
7	Warm-water explorer	03 · Cooling	Your setpoint	45 °C
8	Cooling selector	03 · Cooling	rule-based	–
9	Bengaluru campus calculator	04 · Bengaluru power & water	People equivalent	4,480
10	A day in Bengaluru	04 · Bengaluru power & water	Power cost this hour	₹2,68,268
11	Load-shift calculator	05 · Every TWh is not the same	Saving	₹8.18 crore
12	Five-factor scorecard	05 · Every TWh is not the same	Score	1.88 → Poor
13	Cooling-tower water balance (50 MW)	06 · Water impact	Consumption WUE	1.45 L/kWh
14	Water you can picture	06 · Water impact	Tanker bill	₹8.8 lakh/month
15	Levelised cost per IT-kWh	07 · Unit economics	Break-even rent	₹13,647/kW-month
16	GPU-hour cost	07 · Unit economics	Break-even at ₹150	96%
17	Green upgrade calculator	08 · Green data centres	Payback	2.3 yrs
18	Maturity self-check	08 · Green data centres	Weakest	L1
19	India map	09 · State policies	Index	7.60 / 10
20	Attractiveness index	09 · State policies	Index	7.60 / 10
21	City ranking (Steps A–E)	10 · Where to build	Score	7.27
22	Forecast to 2035	11 · Forecast to 2035	Integrated	4,663 MW
23	Power vs water	12 · Water & energy	Liquid + dry coolers	3.3–3.8
24	Hourly profile	12 · Water & energy	40 °C	1.44 · 1.60 L/kWh → 67.1 kL/h at 42 MW
25	Annual hours	12 · Water & energy	Annual WUE	shown in the chart readout
26	Tower water balance	12 · Water & energy	Tower WUE	2.10 L/kWh

27	Site calculator	12 · Water & energy	Contract demand	`76 MVA`
28	Two buildings	13 · AI vs conventional	AI CO_2	`560 kt`
29	Rack density	13 · AI vs conventional	Equivalent racks	`15`
30	Per prompt	13 · AI vs conventional	Drops in 0.26 mL	`≈ 5 drops`
31	Going dry	13 · AI vs conventional	Break-even grid water	`9.5 L/kWh`
32	Hourly linear programme	14 · Least-water power	₹ per kL saved	`₹85`
33	Price of water saved	14 · Least-water power	Near-zero water	`₹249/kL > ₹99 ✗`
34	Hub ranking	15 · Recommendations	Score	`3.61 / 5`
35	Design check	15 · Recommendations	CUE	`fail ✗`
36	How to read the policy-tools grid	15 · Recommendations	rule-based	—

Overview

OVERVIEW

1. Power flow schematic

INPUTS TAKEN FROM THE SLIDERS AND SETTINGS

SETTING	VALUE USED
Power Usage Effectiveness (total facility energy ÷ IT energy)	1.54
PUE presets	Global avg 1.54

FORMULAS

1. IT share (%) = 100 ÷ PUE
2. Overhead (%) = 100 − IT share
3. Cooling = 70% of overhead · Power-chain losses = 22% · Other (lighting, offices) = 8%
4. Overhead energy (GWh/yr) = 20 MW × 0.70 load × 8.76 × (PUE − 1)
5. Overhead cost (₹ crore) = overhead GWh × ₹8/kWh × 0.1

WHAT EACH EQUATION DOES, AND WHY IT TAKES THIS FORM

- PUE is defined as facility energy ÷ IT energy, so the share reaching IT is simply 1 ÷ PUE.
- Overhead is everything else; the 70 / 22 / 8 split is the typical breakdown of non-IT energy in an air-cooled hall (cooling dominates).
- 20 MW × 70% load is the series' reference colocation hall; (PUE − 1) is the extra energy per unit of IT energy.
- × 8.76 turns MW into GWh a year (8,760 hours ÷ 1,000); × ₹8/kWh × 0.1 turns GWh into ₹ crore (1 GWh × ₹1/kWh = ₹0.1 crore).

ASSUMED VALUES AND THE RATIONALE FOR EACH

VALUE	USED	WHY THIS VALUE (SOURCE)	BASIS
Reference site	20 MW IT at 70% load	A mid-size Indian colocation hall, the most common facility type today (Report 4 central case). (Report 4)	Series assumption
Overhead split	70% cooling / 22% power-chain losses / 8% other	Typical breakdown of non-IT energy in air-cooled facilities; cooling dominates in a hot climate. (Report 2 synthesis of PUE ranges and IEA component shares)	Estimate
Tariff	₹8/kWh landed	A round, landed HT industrial price including demand charges, for illustration only. (Series)	Series assumption

FURTHER NOTES FROM THE ATLAS

- Overhead split 70/22/8 is a series assumption for a typical air-cooled site.
- The example hall is 20 MW of IT at 70% average load, buying power at ₹8/kWh.
- 8.76 = 8,760 hours ÷ 1,000 (MW·h → GWh).

WORKED CALCULATION WITH THE SLIDER VALUES ABOVE

STEP	CALCULATION	RESULT
IT share	100 ÷ 1.54	**64.9%**
Overhead	100 − 64.9	**35.1%**
Cooling / losses / other	35.1% × 0.70 / 0.22 / 0.08	**24.5% / 7.7% / 2.8%**

Overhead energy	`20 × 0.70 × 8.76 × (1.54 − 1)`	**66.2 GWh/yr**
Overhead cost	`66.2 × 8 × 0.1`	**₹53.0 crore/yr**

▶ Build a data centre

▶ BUILD A DATA CENTRE

2. Simulator model

INPUTS TAKEN FROM THE SLIDERS AND SETTINGS

SETTING	VALUE USED
IT capacity	20 MW
Average IT load (utilisation)	70%
Apply the state's published incentives	on
Include GPUs and network in capex	on
City	Bengaluru · Karnataka
Workload	Colocation
Cooling	Cooling towers
Power	Grid only
Water source	Municipal

FORMULAS

1. 1 · Efficiency
2. CRF(r, n) = $r(1 + r)^n \div ((1 + r)^n - 1)$
3. PUE = max(1.03, 1 + power-chain losses + other loads + cooling overhead)
4. Losses / other: colocation 0.10 / 0.03 · hyperscale 0.08 / 0.02 · AI 0.08 / 0.02
5. Cooling overhead, colo & hyperscale: towers 0.30 + 0.012(WB − 23) · adiabatic 0.27 + 0.01(DB − 35) · air chillers 0.42 + 0.015(DB − 35) · liquid 0.12
6. Cooling overhead, AI (liquid in hall): towers 0.10 + 0.004(WB − 23) · adiabatic 0.09 + 0.004(DB − 35) · air chillers 0.16 + 0.006(DB − 35) · liquid 0.09
7. Liquid: +0.02 if DB + 7 > 45
8. WUE (L/kWh): towers 1.8 × (1 + 0.03(WB − 23)) · adiabatic max(0.1, 0.4 × (1 + 0.05(DB − 35))) · air chillers 0.02 · liquid 0.03 (0.1 if DB + 7 > 45)
9. 2 · Energy and grid connection
10. IT energy (MU) = MW × utilisation × 8.76 · facility energy (MU) = IT × PUE
11. Grid connection (MVA) = MW × (PUE + 0.12) ÷ 0.95 → ≤20 MVA 66 kV · ≤55 110 kV · ≤250 220 kV dual feed · above 400 kV
12. Homes equivalent = facility MU × 10^6 ÷ 1,800 kWh · share of Karnataka = facility MU ÷ 93,000
13. 3 · Power price
14. Clean share: grid 0 · 60% renewable 0.6 · hourly-matched 0.9
15. RE price = city open-access price; hourly-matched = max(city price + 0.6, 6.4)
16. Blended price (₹/kWh) = clean share × RE price + (1 − clean share) × grid tariff
17. 4 · Capex (₹ crore)
18. Facility = MW × (workload ₹ cr/MW + cooling adder) [colo 56.1 · hyperscale 39.2 · AI 72.5; adder towers 0 · adiabatic 0.5 · air chillers 1 · liquid 3 (0 for AI)]
19. Land = MW × acres/MW × city land price [acres/MW: colo 0.25 · hyperscale 0.4 · AI 0.3]
20. GPUs (AI, if included) = MW × ₹266 cr · total capex = facility + land + GPUs
21. 5 · Annual cost and unit costs (₹ crore/yr)
22. Capital charge = (facility + land) × CRF(11%, 20) · GPU charge = GPU capex × CRF(11%, 5)
23. Electricity = facility MU × blended price × 0.1
24. Demand charge = MVA × 1,000 × ₹/kVA-month × 12 ÷ 10^7
25. O&M = MW × (colo 2.25 · hyperscale 1.17 · AI 3.29)
26. Water cost = on-site ML × 1,000 × ₹/kL ÷ 10^7 [municipal city rate · groundwater ₹20 · treated ₹40]
27. Annual cost = capital charge + electricity + demand + O&M + water − incentives

28. Cost per IT-kWh (₹) = annual cost ÷ IT MU × 10
29. Break-even rent (₹/kW-month) = annual cost × 10^7 ÷ (MW × 1,000 × 12)
30. GPU-hour (₹) = (annual cost + GPU charge) × 10^7 ÷ (MW × 425 GPUs × 8,760 × utilisation) · US$ = ₹ ÷ 90
31. 6 · Incentives (₹ crore/yr)
32. = facility MU × 0.1 × (₹/unit subsidy + 0.075 × tariff if duty waived) + bonus × CRF(11%, 20) if PUE ≤ 1.20
33. Karnataka: duty waiver + ₹20 cr bonus · Maharashtra: ₹1/unit + duty waiver, only if clean share ≥ 51% · UP: duty waiver · TN, Telangana, AP: none
34. 7 · Water
35. On-site (ML/yr) = IT MU × WUE × RO factor [1 ÷ 0.75 ≈ 1.33 for treated wastewater, else 1] · MLD = ML ÷ 365
36. Power stations (ML/yr) = facility MU × (1 − clean) × grid L/kWh + facility MU × clean × 0.05
37. People equivalent = MLD × 10^6 ÷ 135 L
38. Scarcity-weighted = on-site × source weight × city stress + power-station × 0.4 [groundwater 1 · municipal 0.7 · treated 0.1]
39. 8 · Carbon
40. CO_2 (t/yr) = facility MU × 1,000 × 0.710 × (1 − clean share) · CUE (kg/IT-kWh) = CO_2 ÷ (IT MU × 1,000)
41. 9 · Ranges
42. Model re-run at low / high: capex ×0.88 / ×1.12 · PUE −0.04 / +0.04 · WUE ×0.7 / ×1.35 · tariff −₹0.50 / +₹0.50 · grid 0.65 / 0.80 t/MWh · GPU price ×0.88 / ×1.55

WHAT EACH EQUATION DOES, AND WHY IT TAKES THIS FORM

- PUE is built bottom-up: 1 (the IT itself) + power-chain losses + other loads + a cooling overhead. The overhead grows with the city's design temperature because chillers work harder in hotter or more humid air; AI racks with liquid in the hall need much less.
- WUE comes from the cooling type: towers evaporate the most, and more when the wet-bulb is high; dry and liquid systems use almost none.
- Capex per MW comes from Report 7 cost stacks; cooling choice adds a cost; land = acres needed per MW × the city's price.
- Annual cost puts everything on a yearly basis: capex via CRF(11%, 20 yrs) like a loan, plus electricity, demand charge, O&M and water, minus incentives. Dividing by IT energy gives ₹ per useful kWh.
- The blended power price weights renewable and grid tariffs by the clean share you choose.
- Water at power stations = facility energy × fossil share × grid L/kWh, because coal plants evaporate water to make each kWh. CO_2 uses the CEA factor 0.71 t/MWh on the fossil share.
- Ranges re-run the whole model with each uncertain input at its low and high value, then report the spread.

ASSUMED VALUES AND THE RATIONALE FOR EACH

VALUE	USED	WHY THIS VALUE (SOURCE)	BASIS
Power-chain losses / other loads	colo 0.10 / 0.03; hyperscale and AI 0.08 / 0.02	Transformer, UPS and distribution losses are ~8–10% of IT load; larger sites use more efficient medium-voltage gear. Lighting, offices and security add 2–3%. (Report 2 power-chain losses; LBNL 2024)	Estimate
Cooling overhead (colo, towers)	0.30 + 0.012 × (WB − 23)	0.30 is the overhead of chillers + towers at Bengaluru's 23 °C design wet-bulb; each extra °C of wet-bulb reduces tower effectiveness, adding ~1.2% of IT energy. (Report 3 cooling ranges, calibrated so Bengaluru colo ≈ PUE 1.43)	Estimate

Cooling overhead (AI, liquid in hall)	`towers 0.10 + 0.004(WB − 23); liquid 0.09`	Direct-to-chip liquid removes most heat at 40–45 °C water, so chillers barely run; overhead falls to about a third of air cooling. (Report 3; vendor W40–W45 guidance)	Estimate
Liquid penalty	`+0.02 if DB + 7 > 45`	A dry cooler supplies water ~7 K above air temperature; above 45 °C supply the loop needs trim cooling. (ASHRAE liquid classes; Report 3 approach temperatures)	Physics
PUE floor	`1.03`	No real facility goes below ~1.03 (fans, pumps, conversion losses); prevents unrealistic results at extreme settings. (Series)	Series assumption
WUE (towers)	`1.8 × (1 + 0.03 × (WB − 23)) L/kWh`	1.8 L/kWh is a typical evaporative WUE in India (NTT DATA evaporative sites 1.4–2.9); higher wet-bulb raises evaporation ~3% per °C. (Report 6; NTT DATA FY22–24)	Estimate
WUE (adiabatic / air chillers / liquid)	`max(0.1, 0.4 × (1 + 0.05(DB − 35))) / 0.02 / 0.03`	Adiabatic pads only run in hot hours, so annual WUE is low but rises with design heat; air chillers use water only for humidification; liquid loops need small top-ups. (Report 6; chapter 12 ranges)	Estimate
Capex per MW	`colo ₹56.1 cr, hyperscale ₹39.2 cr, AI ₹72.5 cr (facility)`	Bottom-up cost stacks (shell, electrical, cooling, backup, fire/BMS, soft costs); colo pays for 2N redundancy and small halls; AI pays for heavy power and liquid plant. (Report 7 Cases A–C; Turner & Townsend Mumbai US$6.64/W)	Estimate
Cooling capex adder	`towers 0, adiabatic ₹0.5, air chillers ₹1, liquid ₹3 cr/MW (0 for AI, already included)`	Relative cost of each heat-rejection system versus a tower plant. (Report 3)	Estimate
Land	`acres/MW: colo 0.25, hyperscale 0.4, AI 0.3; city price ₹3–20 cr/acre`	Multi-storey colos need least land; hyperscale campuses spread out; prices are indicative industrial land rates by city (Mumbai highest). (Report 7; Report 10 land data)	Estimate
GPUs	`₹266 cr per MW of IT`	~590 GB300 NVL72 racks for 100 MW at ~US$4.2 m each, plus fabric and storage, ÷ 100 MW. (Report 7 Case C (GB300 price contested))	Estimate
WACC and life	`11% over 20 years (GPUs: 5 years)`	A typical Indian infrastructure cost of capital; buildings last ~20 years, GPUs are obsolete in ~5. (Report 7)	Series assumption
O&M	`colo ₹2.25, hyperscale ₹1.17, AI ₹3.29 cr/MW/yr`	Maintenance contracts, staff, insurance and overheads per MW. (Report 7 opex build-up)	Estimate

Tariffs and demand charges	e.g. Bengaluru ₹6.70/kWh + ₹365/kVA-month	State HT industrial tariffs (FY2025-26 orders); Noida and Vizag not verified (typical ₹7–8). (SERC tariff orders; Report 4, 9)	Reported
Renewable price	city open-access price; hourly-matched = max(price + 0.6, 6.4)	Open-access solar/wind lands at ~₹5.5–6.2/kWh; round-the-clock supply needs storage, adding ~₹0.6. (Report 8; chapter 14 model)	Estimate
Contract demand	MVA = MW × (PUE + 0.12) ÷ 0.95	Grid must carry the hot-hour peak (average PUE + ~0.12) at a 0.95 power factor. (Report 4; KERC conditions of supply)	Estimate
Grid water	Karnataka 1.5, others 2.2 L/kWh	Water evaporated at coal plants per kWh delivered, lower in Karnataka with more hydro, wind and solar. (Report 6 series estimate)	Estimate
Water price	municipal ₹99 (Bengaluru) / ₹80 elsewhere; groundwater ₹20; treated ₹40 per kL	BWSSB bulk industrial tariff; groundwater cost of pumping and treatment; treated sewage water contract price. (BWSSB tariff; Report 6)	Estimate
Incentives	duty waiver = 7.5% of energy charge; Karnataka ₹20 cr bonus if PUE ≤ 1.20; Maharashtra ₹1/unit if ≥51% RE	Published state incentives (Report 9); electricity duty modelled at a typical 7.5%. (State policies)	Reported
Ranges	capex ±12%, PUE ±0.04, WUE ×0.7/×1.35, tariff ±₹0.50, grid 0.65/0.80, GPU ×0.88/×1.55	The main uncertainties: construction cost spread, climate year-to-year, measurement basis, tariff revisions, future grid mix, contested GPU prices. (Series)	Series assumption

FURTHER NOTES FROM THE ATLAS

- Units: MU = million kWh; ₹ cr = ₹ crore; ML = million litres. ×0.1 turns MU × ₹/kWh into ₹ crore; ×10 turns ₹ crore per MU into ₹/kWh.
- DB / WB = city design dry-bulb / wet-bulb (°C).
- City inputs (DB/WB °C · land ₹ cr/acre · tariff ₹/kWh · demand ₹/kVA-month · RE ₹/kWh · grid water L/kWh · water stress): Bengaluru 35/23 · 8 · 6.70 · 365 · 6.2 · 1.5 · 1 — Mumbai 35/28 · 20 · 7.27 · 560 · 5.8 · 2.2 · 0.5 — Chennai 38/29 · 6 · 7.50 · 608 · 5.9 · 2.2 · 0.9 — Hyderabad 41/26 · 8 · 6.65 · 500 · 6.1 · 2.2 · 0.7 — Noida 45/29 · 10 · 7.40 · 450 · 6.0 · 2.2 · 0.85 — Vizag 36/28 · 3 · 7.20 · 475 · 5.5 · 2.2 · 0.6.
- Municipal water: ₹99/kL Bengaluru, ₹80/kL elsewhere. Noida and Vizag tariffs not verified (typical ₹7–8).
- Evidence for every input is in "Assumptions behind the simulator" at the end of this section.

WORKED CALCULATION WITH THE SLIDER VALUES ABOVE

STEP	CALCULATION	RESULT
City	design dry-bulb / wet-bulb	**35 / 23 °C**
PUE	1 + 0.1 losses + 0.03 other + 0.300 cooling	**1.43**
WUE	tower at 35/23 °C	**1.80 L/kWh**
IT energy	20 MW × 70% × 8.76	**122.6 MU**
Facility energy	122.6 × 1.43	**175.4 MU**

Grid connection	20 × (1.43 + 0.12) ÷ 0.95	**33 MVA**
Facility capex	20 MW × (₹56.1 + ₹0.0 cooling) cr	**₹1,122 cr**
Land	20 × 0.25 acre/MW × ₹8 cr	**₹40.0 cr**
Capital charge	(1,122 + 40.0) × CRF(11%,20) = × 0.1256	**₹145.9 cr/yr**
Blended price	0% × ₹6.20 + 100% × ₹6.70	**₹6.70/kWh**
Electricity	175.4 MU × ₹6.70 × 0.1	**₹117.5 cr/yr**
Demand charge	32.6 MVA × 1,000 × ₹365 × 12 ÷ 10^7	**₹14.3 cr/yr**
O&M	20 × ₹2.25 cr	**₹45.0 cr/yr**
Water cost	220.8 ML × 1,000 × ₹99/kL ÷ 10^7	**₹2.19 cr/yr**
Incentives	state rules (chapter 9)	**−₹8.8 cr/yr**
Annual cost	145.9 + 179.0 − 8.8	**₹316.1 cr/yr**
Cost per IT-kWh	316.1 ÷ 122.6 × 10	**₹25.8**
Break-even rent	316.1 × 10^7 ÷ (20 × 1,000 × 12)	**₹13,170/kW-month**
On-site water	122.6 × 1.80	**220.8 ML/yr**
Power-station water	175.4 × 100% × 1.5 + clean × 0.05	**263.1 ML/yr**
CO_2	175.4 × 1,000 × 0.71 × 1.00	**1,24,516 t/yr**

▶ BUILD A DATA CENTRE

3. Flow diagrams

On the atlas panel: *Follow the flow. 175 MU a year enters; 70% reaches the servers. Every unit ends as heat.*

INPUTS TAKEN FROM THE SLIDERS AND SETTINGS

SETTING	VALUE USED
Flow	Power

FORMULAS

1. Power: losses = IT × loss factor; other = IT × other factor; cooling = facility − IT − losses − other; all energy leaves as heat
2. Water: make-up = on-site ÷ 1.33 (if RO); RO reject = on-site − make-up; evaporated share: towers 70%, adiabatic 90%, air chillers 50%, liquid 85%; rest is blowdown. Power-station water is counted as evaporated.
3. Money: facility capex × CRF(11%, 20) split shell 15%, electrical 39%, cooling 18%, backup 8%, fire/BMS/network 5%, soft costs 15%; land × CRF; GPUs × CRF(11%, 5); plus electricity, demand, O&M and water

WHAT EACH EQUATION DOES, AND WHY IT TAKES THIS FORM

- Energy is conserved: everything bought ends up as heat. So cooling energy can be found as what is left after IT, losses and other loads.
- Water make-up splits into evaporation (lost to air) and blowdown (returned); each cooling type evaporates a typical share. RO reject is extra withdrawal that returns to drain.
- Money flows split the capital charge by the Report 7 colocation cost stack so you can see which systems drive cost.

ASSUMED VALUES AND THE RATIONALE FOR EACH

VALUE	USED	WHY THIS VALUE (SOURCE)	BASIS

Evaporated share of make-up	towers 70%, adiabatic 90%, air chillers 50%, liquid 85%	The rest of the make-up water leaves as blowdown; adiabatic pads evaporate almost all of their water. (Report 6)	Estimate
Capex split	shell 15%, electrical 39%, cooling 18%, backup 8%, fire/BMS/network 5%, soft 15%	The Report 7 colocation cost stack, used to colour the money flow. (Report 7)	Estimate

FURTHER NOTES FROM THE ATLAS

- Band widths are proportional to annual values; very small flows are drawn at a minimum visible width.
- Capex split follows the Report 7 colocation cost stack.

WORKED CALCULATION WITH THE SLIDER VALUES ABOVE

STEP	CALCULATION	RESULT
Losses	122.6 × 0.1	**12.3 MU**
Other loads	122.6 × 0.03	**3.7 MU**
Cooling	175.4 − 122.6 − 12.3 − 3.7	**36.8 MU (21%)**
Water make-up	220.8 ÷ 1.00	**220.8 ML**

01 · Types of data centre

01 · TYPES OF DATA CENTRE

4. Tier downtime

On the atlas panel: *What a Tier buys you. Each square is one of 120 slices of a year; orange slices show allowed downtime, magnified 60× so it is visible.*

INPUTS TAKEN FROM THE SLIDERS AND SETTINGS

SETTING	VALUE USED
Tier	Tier III

FORMULAS

1. Downtime (hours/yr) = (1 − availability) × 8,760
2. Red squares = downtime ÷ 8,760 × 120 squares × 60

WHAT EACH EQUATION DOES, AND WHY IT TAKES THIS FORM

- Availability is the share of the 8,760 hours in a year the site is designed to be up, so downtime is the remainder: (1 − availability) × 8,760.
- Each of the 120 squares stands for 73 hours. Real Tier III/IV downtime would be a fraction of one square, so the clock magnifies it 60× to make it visible.

ASSUMED VALUES AND THE RATIONALE FOR EACH

VALUE	USED	WHY THIS VALUE (SOURCE)	BASIS
Availability	Tier I 99.671%, II 99.741%, III 99.982%, IV 99.995%	Uptime Institute design targets for each Tier. (Uptime Institute)	Reported
Magnification	60×	Real Tier III/IV downtime is a fraction of one of 120 squares; 60× makes it visible. (Display choice)	Convention

FURTHER NOTES FROM THE ATLAS

- Availability figures are the Uptime Institute design targets for Tiers I–IV.
- The clock exaggerates downtime 60× so that Tier III and IV are visible at all.

WORKED CALCULATION WITH THE SLIDER VALUES ABOVE

STEP	CALCULATION	RESULT
Tier III downtime	(1 − 99.982%) × 8,760	**1.58 h/yr**
Red squares	1.58 ÷ 8,760 × 120 × 60	**1**

01 · TYPES OF DATA CENTRE

5. Type finder

On the atlas panel: *Type finder. Answer four questions and get the type the series would recommend.*

INPUTS TAKEN FROM THE SLIDERS AND SETTINGS

SETTING	VALUE USED
How much IT load?	1–20 MW
GPUs / AI training?	No
Must be close to users (<20 ms)?	No
Regulated data / full control?	Yes

FORMULAS

1. Rule-based, no arithmetic. Inputs: size band, AI workload, latency-critical, regulated data.

WHAT EACH EQUATION DOES, AND WHY IT TAKES THIS FORM

- The type finder has no arithmetic: it follows the decision order the series recommends.
- Size first, because scale sets the business model (≥100 MW only works as hyperscale; under 1 MW is edge or public cloud).
- Then AI workload (changes cooling and redundancy), latency (pushes small sites to the edge) and regulation (pushes them into Tier III colo racks).

ASSUMED VALUES AND THE RATIONALE FOR EACH

VALUE	USED	WHY THIS VALUE (SOURCE)	BASIS
Size bands	≥100 MW, 20–100, 1–20, <1 MW	Breakpoints where the business model changes (hyperscale, wholesale, retail colo, edge). (Report 1)	Series assumption
GPU rent vs own	break-even ~55% utilisation for an H100	From the chapter 7 GPU-hour model at market prices. (Report 7)	Estimate

FURTHER NOTES FROM THE ATLAS

- ≥100 MW → hyperscale (AI: power-first campus, liquid at W40–W45).
- 20–100 MW → wholesale / build-to-suit (AI: wholesale with AI halls or GPU cloud).
- 1–20 MW → retail colo or managed cloud (AI: GPU-as-a-service; renting beats owning an H100 below ~55% utilisation).
- <1 MW → edge if latency-critical; colo racks if regulated; otherwise public cloud.
- Reliability: regulated → Tier III (Tier IV only for trading or critical national systems); AI training can run N+1.

WORKED RESULT

Rule-based: the result is the branch of the rules above that matches the inputs; there is no arithmetic.

02 · AI vs conventional

02 · AI VS CONVENTIONAL

6. Rack inspector

On the atlas panel: *Rack inspector. Slide the rack's power draw and see what it forces on the building.*

INPUTS TAKEN FROM THE SLIDERS AND SETTINGS

SETTING	VALUE USED
Rack power	6 kW

FORMULAS

1. Rack power (kW) = round(3 × 200^(slider ÷ 100)) — a log slider from 3 to 600 kW
2. Racks per MW of IT = floor(1,000 ÷ kW)
3. Homes' yearly use = kW × 8,760 h × 0.8 load ÷ 1,800 kWh

WHAT EACH EQUATION DOES, AND WHY IT TAKES THIS FORM

- The slider is logarithmic (3 × 200^(s/100)) because rack power spans 3 to 600 kW: equal slider moves give equal ratios, so a 10 kW and a 300 kW rack are both easy to reach.
- A MW of IT is 1,000 kW, so 1,000 ÷ kW gives how many such racks one MW feeds.
- A rack runs at about 80% of its rating on average; 1,800 kWh is a Bengaluru household's yearly use (Report 4), which makes the number tangible.

ASSUMED VALUES AND THE RATIONALE FOR EACH

VALUE	USED	WHY THIS VALUE (SOURCE)	BASIS
Slider range	3–600 kW, logarithmic	From an enterprise rack to NVIDIA's 2027 Kyber rack; log scale so both ends are usable. (Vendor roadmaps)	Convention
Average draw	80% of rating	Racks rarely run at nameplate around the clock. (Series)	Series assumption
Home	1,800 kWh/yr	A Bengaluru household's yearly use. (Report 4)	Estimate
Class bands	15 / 40 / 150 kW	Practical limits of air with containment, rear-door coils, and direct-to-chip liquid. (Report 3)	Estimate

FURTHER NOTES FROM THE ATLAS

- Class bands: ≤15 kW conventional air; 15–40 kW high-density (RDHx/in-row); 40–150 kW AI direct-to-chip; >150 kW next-gen, 100% liquid.
- Liquid share of heat per band: 0%, 40%, 90%, 100% (indicative).
- A Bengaluru home uses ~1,800 kWh a year (Report 4).

WORKED CALCULATION WITH THE SLIDER VALUES ABOVE

STEP	CALCULATION	RESULT
Rack power	round(3 × 200^(12 ÷ 100))	**6 kW**
Racks per MW	floor(1,000 ÷ 6)	**166**
Homes' yearly use	6 × 8,760 × 0.8 ÷ 1,800	**23 homes**
Cooling class	6 kW vs 15 / 40 / 150 kW bands	**Air**

03 · Cooling

03 · COOLING

7. Warm-water explorer

On the atlas panel: *Warm-water explorer. Set the facility water temperature. Each city card shows what peak-day heat rejection that temperature needs (indicative design temperatures; dry cooler approach 7 K, tower 4 K).*

INPUTS TAKEN FROM THE SLIDERS AND SETTINGS

SETTING	VALUE USED
Facility water supply	45 °C

FORMULAS

1. Dry cooler can supply = design dry-bulb + 7 K
2. Adiabatic cooler can supply = wet-bulb + 0.3 × (dry-bulb − wet-bulb) + 7 K
3. Cooling tower can supply = wet-bulb + 4 K
4. City verdict = the first option whose supply temperature ≤ your water setpoint; otherwise a chiller is needed at peak

WHAT EACH EQUATION DOES, AND WHY IT TAKES THIS FORM

- Heat can only flow from hot to cold, so each heat-rejection device can cool water only to a few degrees above the air it uses: the "approach".
- A dry cooler uses dry air: best supply ≈ design dry-bulb + 7 K.
- A cooling tower evaporates water, so it approaches the wet-bulb: ≈ wet-bulb + 4 K.
- An adiabatic cooler sprays the incoming air and gets it most of the way to wet-bulb, leaving about 30% of the gap, then adds the dry-cooler approach.
- If the supply temperature an option can deliver is at or below your water setpoint, that option works on the hottest design day without a chiller.

ASSUMED VALUES AND THE RATIONALE FOR EACH

VALUE	USED	WHY THIS VALUE (SOURCE)	BASIS
Dry-cooler approach	7 K	Typical design approach between leaving water and ambient air for a dry cooler. (Report 3)	Estimate
Tower approach	4 K over wet-bulb	Typical cooling-tower approach. (Report 3)	Estimate
Adiabatic effectiveness	leaves 30% of the wet-bulb depression	Evaporative pre-cooling pads achieve ~70% effectiveness. (Report 3)	Estimate
City design temperatures	e.g. Bengaluru 35/23 °C, Delhi 45/29 °C	Indicative 0.4% summer design conditions (hottest ~35 hours a year), not station data. (Report 3; ISHRAE-style values)	Estimate

FURTHER NOTES FROM THE ATLAS

- Approach temperatures (7 K dry, 4 K tower, 30% of wet-bulb depression for adiabatic) are typical design values.
- City design temperatures are indicative 0.4% summer conditions, not station data.
- ASHRAE liquid classes: W17, W27, W32, W40, W45, W+.

WORKED CALCULATION WITH THE SLIDER VALUES ABOVE

STEP	CALCULATION	RESULT
Bengaluru	dry 35+7=42 · adiabatic 23+0.3×12+7=33.6 · tower 23+4=27	**Dry only**
Hyderabad	dry 41+7=48 · adiabatic 26+0.3×15+7=37.5 · tower 26+4=30	**Adiabatic on peak**

Pune	dry 39+7=46 · adiabatic 25+0.3×14+7=36.2 · tower 25+4=29	**Adiabatic on peak**
Mumbai	dry 35+7=42 · adiabatic 28+0.3×7+7=37.1 · tower 28+4=32	**Dry only**
Chennai	dry 38+7=45 · adiabatic 29+0.3×9+7=38.7 · tower 29+4=33	**Dry only**
Visakhapatnam	dry 36+7=43 · adiabatic 28+0.3×8+7=37.4 · tower 28+4=32	**Dry only**
Delhi NCR	dry 45+7=52 · adiabatic 29+0.3×16+7=40.8 · tower 29+4=33	**Adiabatic on peak**
Your setpoint	compared with each supply temperature	**45 °C**

03 · COOLING

8. Cooling selector

On the atlas panel: *Cooling selector. Describe your site; the selector recommends a configuration and highlights it below.*

INPUTS TAKEN FROM THE SLIDERS AND SETTINGS

SETTING	VALUE USED
Average rack density	10 kW
Water stress	High (Bengaluru)
Climate	Deccan plateau

FORMULAS

1. Rule-based on three inputs: rack density, water stress, climate.
2. Room: >40 kW → direct-to-chip liquid; 15–40 kW → rear-door / in-row; ≤15 kW → air with containment
3. Heat rejection: high water stress → dry coolers; medium → adiabatic / hybrid; low → towers on treated wastewater
4. Coastal + >15 kW → seawater heat exchange added as an option

WHAT EACH EQUATION DOES, AND WHY IT TAKES THIS FORM

- Rack density decides the room: air can carry away ~15 kW per rack with containment, rear-door coils reach ~40 kW, beyond that only liquid at the chip works.
- Water stress decides heat rejection: where water is scarce, use dry coolers (no evaporation) and accept a little more electricity.
- Coastal dense sites get seawater as an option because the sea is a large, cheap heat sink, but it needs an environmental study.

ASSUMED VALUES AND THE RATIONALE FOR EACH

VALUE	USED	WHY THIS VALUE (SOURCE)	BASIS
Room thresholds	15 kW / 40 kW	Above ~15 kW air needs rear-door or in-row coils; above ~40 kW only liquid at the chip works. (Report 3)	Estimate
Expected WUE	dry/DLC 0–0.2, adiabatic 0.1–0.7, towers 1.4–2.0 L/kWh	Typical annual ranges in Indian climates. (Report 3, 6)	Estimate

FURTHER NOTES FROM THE ATLAS

- Expected WUE ranges (L/kWh): dry/DLC 0–0.2, adiabatic 0.1–0.7, towers 1.4–2.0.
- Practical rack ceilings (kW): perimeter air 15, containment 30, active RDHx 70, direct-to-chip and immersion 250.

WORKED RESULT

Rule-based: the result is the branch of the rules above that matches the inputs; there is no arithmetic.

04 · Bengaluru power & water

04 · BENGALURU POWER & WATER

9. Bengaluru campus calculator

On the atlas panel: *Campus calculator. The presets are the report's three central cases. Change anything.*

INPUTS TAKEN FROM THE SLIDERS AND SETTINGS

SETTING	VALUE USED
IT capacity (MW)	20 MW
Average IT load	70%
PUE	1.45
On-site WUE (L/kWh IT)	1.80
Grid carbon (t/MWh)	0.71
Power-station water (L/kWh)	1.5
Presets	20 MW colo, air

FORMULAS

1. IT energy (MU/yr) = MW × utilisation × 8.76
2. Facility energy (MU/yr) = IT energy × PUE
3. Households = facility kWh ÷ 1,800 kWh
4. Share of Karnataka demand = facility MU ÷ 93,000 MU
5. Share of household use = facility MU ÷ 16,089 MU
6. Contract demand (MVA) = MW × (PUE + 0.12) ÷ 0.95
7. CO_2 (t/yr) = facility MU × emission factor × 1,000
8. On-site water (MLD) = IT MU × WUE ÷ 365
9. Power-station water (MLD) = facility MU × grid water (L/kWh) ÷ 365
10. People equivalent = on-site MLD × 10^6 ÷ 135 L

WHAT EACH EQUATION DOES, AND WHY IT TAKES THIS FORM

- MW × 8,760 hours = MWh; ÷ 1,000 = GWh, which Indian utilities call MU (million units). Utilisation is applied because halls rarely run at full nameplate.
- Facility energy = IT energy × PUE, by the definition of PUE.
- Households (1,800 kWh/yr) and Karnataka demand (93,000 MU) give scale; they are comparisons, not part of the physics.
- Contract demand: the grid must be sized for the hot-hour peak, so 0.12 is added to the average PUE; ÷ 0.95 converts real power (MW) to apparent power (MVA) at a 0.95 power factor.
- CO_2: MU × t/MWh × 1,000 = tonnes. Water: MU × L/kWh = million litres (10^6 kWh × 1 L), ÷ 365 for a daily figure. 135 L is the CPHEEO daily need per person.

ASSUMED VALUES AND THE RATIONALE FOR EACH

VALUE	USED	WHY THIS VALUE (SOURCE)	BASIS
Household use	1,800 kWh/yr	Average Bengaluru household consumption, used only to give a sense of scale. (Report 4 (BESCOM data))	Estimate
Karnataka demand	93,000 MU/yr; household use 16,089 MU	State energy requirement and domestic consumption. (CEA / KERC)	Reported

Peak margin and power factor	+0.12 on PUE; 0.95	Hot-hour cooling plus auxiliaries above the annual average; typical power factor with correction. (Report 4)	Estimate
Person's water	135 L/day	CPHEEO urban water supply norm. (CPHEEO)	Reported
BWSSB supply / East groundwater	1,935 MLD / ~45 MLD	City piped supply; sustainable groundwater in Bangalore East used for context. (BWSSB; Report 4)	Estimate
Presets	20 MW colo, 50 MW hyperscale, 100 MW AI	Report 4's three central cases. (Report 4)	Series assumption

FURTHER NOTES FROM THE ATLAS

- 0.12 adds cooling and auxiliary peak above average PUE; 0.95 is the power factor.
- Voltage: ≤20 MVA 66 kV; ≤55 MVA 110 kV; ≤250 MVA 220 kV dual feed; above that 400 kV.
- BWSSB supply 1,935 MLD; Bangalore East sustainable groundwater ~45 MLD (series estimate).
- 1 MU = 1 million kWh = 1 GWh.

WORKED CALCULATION WITH THE SLIDER VALUES ABOVE

STEP	CALCULATION	RESULT
IT energy	20 × 70% × 8.76	**122.6 MU**
Facility energy	122.6 × 1.45	**177.8 MU**
Households	177.8 × 10^6 ÷ 1,800	**98,793**
Share of Karnataka	177.8 ÷ 93,000	**0.19%**
Contract demand	20 × (1.45 + 0.12) ÷ 0.95	**33 MVA**
CO_2	177.8 × 0.71 × 1,000	**1,26,258 t**
On-site water	122.6 × 1.80 ÷ 365	**0.60 MLD**
Power-station water	177.8 × 1.5 ÷ 365	**0.73 MLD**
People equivalent	0.60 × 10^6 ÷ 135	**4,480**

04 · BENGALURU POWER & WATER

10. A day in Bengaluru

On the atlas panel: *A day in Bengaluru, hour by hour. Scrub through 24 hours. Grid carbon, the power tariff, outdoor heat, how the cooling plant runs and the IT load all change through the day, and so do the cost, carbon and water of each u*

INPUTS TAKEN FROM THE SLIDERS AND SETTINGS

SETTING	VALUE USED
IT capacity	50 MW
Flexible load moved from evening peak (18–23 h) to solar hours (10–15 h)	0%
Hour	15:00
Season	April heat
Workload	Colo (steady)

Cooling	Cooling towers

FORMULAS

1. Dry-bulb(h) = low + (high − low) × shape(h); shape = 0 at 06:00, 1 at 15:00 (cosine)
2. Wet-bulb(h) = same curve between seasonal wet-bulb low and high
3. PUE(h) = 1 + losses + other + cooling overhead at that hour's temperatures (simulator formulas)
4. Free cooling: air chillers when DB < 18 °C; towers when WB < 14 °C (overhead × 0.6); adiabatic dry mode when DB ≤ 30 °C (× 0.85); liquid dry-only when DB + 7 ≤ 40
5. Facility MW(h) = IT capacity × load(h) × PUE(h)
6. Load shift: load at 18–23 h × (1 − shift%); the removed energy is spread evenly over 10–15 h
7. Water (kL/h) = IT MW × 1,000 × WUE(h) ÷ 1,000; WUE: towers 1.8 × (1 + 0.03(WB − 23)), adiabatic 1.2 in wet hours, liquid 0.5 in trim hours
8. CO_2 (t/h) = facility MW × grid g/kWh ÷ 1,000 · cost = facility kWh × tariff(h)
9. Tariff(h) = ₹6.70 × 1.2 (18–22 h) · × 0.8 (10–16 h) · × 1 otherwise

WHAT EACH EQUATION DOES, AND WHY IT TAKES THIS FORM

- Outdoor temperature follows a daily cycle: coolest near 06:00, hottest near 15:00. A cosine curve between the season's low and high reproduces that shape.
- PUE is recomputed every hour because cooling overhead depends on that hour's air: in cool hours equipment can switch to free cooling, which cuts the overhead.
- Facility MW = IT MW × PUE; water = IT MWh × WUE; CO_2 = facility MWh × that hour's grid intensity; cost = facility kWh × that hour's time-of-day tariff (+20% evening, −20% solar hours).
- Load shifting removes part of the 18:00–23:00 load and spreads the same energy over 10:00–15:00, when the grid is cleaner and cheaper.

ASSUMED VALUES AND THE RATIONALE FOR EACH

VALUE	USED	WHY THIS VALUE (SOURCE)	BASIS
Season temperatures	April 22–35 °C (WB 19–23); July 20–28 (19–22); January 15–28 (12–17)	Typical Bengaluru daily ranges by season. (Report 4 / IMD normals (approx.))	Estimate
Daily shape	cosine, low 06:00, high 15:00	Standard daily temperature cycle used in building energy models. (Engineering convention)	Convention
Free-cooling rules	air chillers below 18 °C; towers below 14 °C WB (×0.6); adiabatic dry ≤30 °C (×0.85)	When outside air is cool enough, compressors can be bypassed. (Report 3)	Estimate
Grid carbon	Report 5 Karnataka hourly curve × 1.05 April, 0.8 July, 1.0 January	Monsoon wind lowers July intensity; hot April raises demand on coal. (Report 5 (illustrative))	Estimate
Time-of-day tariff	₹6.70 × 1.2 (18–22 h), × 0.8 (10–16 h)	Karnataka ToD structure, simplified to ±20%. (KERC ToD; Report 5)	Estimate
Load profiles	colo 70% flat; training 85% flat; inference 36–97% diurnal	Typical utilisation patterns by workload. (Report 2)	Estimate

FURTHER NOTES FROM THE ATLAS

- Seasons (°C): April 22–35, WB 19–23; July 20–28, WB 19–22; January 15–28, WB 12–17.
- Grid carbon = Report 5 Karnataka curve × 1.05 April, 0.8 July (monsoon wind), 1.0 January.
- Loads: colo 70% flat; AI training 85% flat; inference follows a diurnal curve (36–97%).

WORKED CALCULATION WITH THE SLIDER VALUES ABOVE

STEP	CALCULATION	RESULT
Hour	15:00, April	**shape = 1.00**
Dry-bulb	22 + (35 − 22) × 1.00	**35.0 °C**
Wet-bulb	19 + (23 − 19) × 1.00	**23.0 °C**
Cooling mode	rule for this cooling type at these temperatures	**Chillers + towers**
PUE	1 + losses + other + cooling overhead	**1.43**
IT load	50 MW × 70%	**35.0 MW**
Facility power	35.0 × 1.43	**50.1 MW**
Water this hour	IT MW × WUE(h)	**63.0 kL**
CO_2 this hour	50.1 × 273 g/kWh ÷ 1,000	**13.7 t**
Power cost this hour	50.1 × 1,000 × ₹5.36	**₹2,68,268**

05 · Every TWh is not the same

11. Load-shift calculator

On the atlas panel: *Load-shift calculator. Move flexible work (training, batch inference, backups) from the evening peak into solar hours. India's ToD rules give ≥20% off in solar hours and ≥20% extra at peak.*

INPUTS TAKEN FROM THE SLIDERS AND SETTINGS

SETTING	VALUE USED
Shiftable load (MW)	20 MW
Hours shifted per day	4 h
Days per year	365
Base tariff (₹/kWh)	₹7.0

FORMULAS

1. Energy moved (MWh/yr) = MW shifted × hours a day × days a year
2. CO_2 avoided (t) = MWh × 0.30 t/MWh
3. ToD spread (₹/kWh) = tariff × 1.2 − tariff × 0.8 = tariff × 0.4
4. Saving (₹ crore) = MWh × 1,000 × spread ÷ 10^7

WHAT EACH EQUATION DOES, AND WHY IT TAKES THIS FORM

- Energy moved = MW shifted × hours a day × days a year.
- CO_2 avoided uses the gap between evening (~0.58 t/MWh) and solar-hour (~0.25 t/MWh) grid intensity on the Karnataka curve: ~0.30 t per MWh moved. It is an average-basis estimate; the true marginal saving could differ.
- Time-of-day tariffs charge about +20% at the evening peak and −20% in solar hours, so moving 1 kWh saves 0.4 × tariff.

ASSUMED VALUES AND THE RATIONALE FOR EACH

VALUE	USED	WHY THIS VALUE (SOURCE)	BASIS
CO_2 per MWh moved	0.30 t	Gap between evening (~0.58) and solar-hour (~0.25) average intensity on the Karnataka curve. Average, not marginal, basis. (Report 5)	Estimate
ToD spread	±20%	Simplified ToD: +20% at evening peak, −20% in solar hours. (Report 5; state ToD rules)	Estimate

FURTHER NOTES FROM THE ATLAS

- 0.30 t/MWh ≈ gap between evening (~0.58) and solar-hour (~0.25) average grid intensity on the Report 5 Karnataka curve. Average basis, not marginal.
- Time-of-day pricing modelled as +20% at peak and −20% in solar hours.

WORKED CALCULATION WITH THE SLIDER VALUES ABOVE

STEP	CALCULATION	RESULT
Energy moved	20 × 4 × 365	**29,200 MWh**
CO_2 avoided	29,200 × 0.30	**8,760 t**
Tariff spread	7.00 × 1.2 − 7.00 × 0.8	**₹2.80/kWh**
Saving	29,200 × 1,000 × 2.80 ÷ 10^7	**₹8.18 crore**

12. Five-factor scorecard

On the atlas panel: *Five-factor scorecard. Score a site or power contract 1–5 on each factor. Weights: carbon 30%, water 25%, source 20%, timing 15%, location 10%.*

INPUTS TAKEN FROM THE SLIDERS AND SETTINGS

SETTING	VALUE USED
Source · Who supplies each hour?	2
Timing · Hourly matched? Flexible load in solar hours?	2
Location · Marginal factor, losses	2
Carbon · tCO_2/MWh	2
Water · Site + source water, scarcity-weighted	1.5
Scorecard preset	Bengaluru, grid only

FORMULAS

1. Score = 0.20 × Source + 0.15 × Timing + 0.10 × Location + 0.30 × Carbon + 0.25 × Water
2. Verdict: ≥4 Leading · ≥3 Improving · below 3 Poor

WHAT EACH EQUATION DOES, AND WHY IT TAKES THIS FORM

- A weighted average turns five 1–5 scores into one 1–5 score. Weights add to 1, so the result stays on the same scale.
- Carbon (30%) and water (25%) are the outcomes that matter; source (20%), timing (15%) and location (10%) are the drivers that produce them, so they weigh less.
- Thresholds: 4+ means most factors are strong (Leading), 3+ means mixed (Improving).

ASSUMED VALUES AND THE RATIONALE FOR EACH

VALUE	USED	WHY THIS VALUE (SOURCE)	BASIS
Weights	carbon 30, water 25, source 20, timing 15, location 10	Outcomes (carbon, water) weigh more than their drivers. (Report 5 framework)	Series assumption
Verdict bands	≥4 Leading, ≥3 Improving	Simple thresholds on a 1–5 scale. (Report 5)	Series assumption

FURTHER NOTES FROM THE ATLAS

- Weights are the Report 5 framework; each factor scored 1–5 by the user.

WORKED CALCULATION WITH THE SLIDER VALUES ABOVE

STEP	CALCULATION	RESULT
Source	2 × 0.20	**0.40**
Timing	2 × 0.15	**0.30**
Location	2 × 0.10	**0.20**
Carbon	2 × 0.30	**0.60**
Water	1.5 × 0.25	**0.38**
Score	sum of the above	**1.88 → Poor**

06 · Water impact

06 · WATER IMPACT

13. Cooling-tower water balance (50 MW)

On the atlas panel: *Cooling-tower water balance, 50 MW hyperscale. Annual ML at 80% load. CoC is how many times water is reused before it is bled off.*

INPUTS TAKEN FROM THE SLIDERS AND SETTINGS

SETTING	VALUE USED
Cycles of concentration	5
Treatment mode	Standard

FORMULAS

1. Evaporation E = 504.6 ML/yr
2. Blowdown B = E ÷ (CoC − 1)
3. Standard: withdrawal = E + B + drift + non-cooling · discharge = B + 0.8 × non-cooling
4. Treated + RO: make-up M = E + B + drift · RO feed = M ÷ 0.75 · reject = feed − M · withdrawal = feed + non-cooling
5. Zero liquid discharge: withdrawal = E + drift + non-cooling · discharge = 0
6. Consumption = withdrawal − discharge
7. WUE (L/kWh) = ML ÷ 350.4 GWh of IT energy

WHAT EACH EQUATION DOES, AND WHY IT TAKES THIS FORM

- Evaporation is set by heat, not by water quality: rejecting heat by evaporation takes ~2,430 kJ per kg of water, so 350.4 GWh of IT heat a year evaporates ~504.6 ML.
- Cycles of concentration (CoC) is how many times the circulating water is concentrated before salts must be bled off. A salt balance gives blowdown B = E ÷ (CoC − 1): higher CoC, less blowdown.
- Withdrawal = everything taken in (evaporation + blowdown + drift + domestic use); discharge = what goes back (blowdown + 80% of domestic use as sewage); consumption = withdrawal − discharge.
- With treated wastewater and RO, the plant recovers 75% of feed, so feed = make-up ÷ 0.75 and the rest is reject. Zero liquid discharge removes discharge entirely, so all withdrawal becomes consumption.
- WUE = ML ÷ GWh, which is the same as L/kWh.

ASSUMED VALUES AND THE RATIONALE FOR EACH

VALUE	USED	WHY THIS VALUE (SOURCE)	BASIS
IT energy	350.4 GWh (50 MW × 80% × 8,760 h)	The Report 6 hyperscale worked case. (Report 6)	Series assumption
Evaporation	504.6 ML/yr (~1.44 L per kWh of heat)	Latent heat ~2,430 kJ/kg at ~30 °C with a typical latent share of heat rejection. (Physics; Report 6)	Physics
Drift / non-cooling use	1.2 ML / 14.6 ML (80% returned as sewage)	Spray loss from towers; domestic and humidification use. (Report 6)	Estimate
RO recovery	75%	Typical recovery when polishing treated wastewater. (Report 6)	Estimate

FURTHER NOTES FROM THE ATLAS

- 50 MW × 80% load = 350.4 GWh IT a year; E = 350.4 × ~1.44 L per kWh of heat.
- Drift 1.2 ML; non-cooling (domestic, humidification) 14.6 ML, 80% returned as sewage.
- RO recovery 75%. ZLD adds 20–70 kWh per m³ of energy.

WORKED CALCULATION WITH THE SLIDER VALUES ABOVE

STEP	CALCULATION	RESULT
Blowdown	`504.6 ÷ (5 − 1)`	**126.2 ML**
Withdrawal	`evaporation + blowdown + drift + domestic (per mode)`	**646.6 ML**
Discharge	`returned flows (per mode)`	**137.8 ML**
Consumption	`646.6 − 137.8`	**508.7 ML**
Withdrawal WUE	`646.6 ÷ 350.4`	**1.85 L/kWh**
Consumption WUE	`508.7 ÷ 350.4`	**1.45 L/kWh**

06 · WATER IMPACT

14. Water you can picture

On the atlas panel: *Water you can picture. Turn million litres into people, tanker trucks and swimming pools, then test the site against a 2024-style summer.*

INPUTS TAKEN FROM THE SLIDERS AND SETTINGS

SETTING	VALUE USED
Cut in city water supply	`30%`
Share of borewells that fail	`47%`
Price per 12,000-litre tanker	`₹1,500`
On-site water storage	`3 ML`
Include water consumed at power stations	`off`
Facility	`20 MW colo · towers`

FORMULAS

1. `MLD = (on-site ML/yr [+ power-station ML/yr]) ÷ 365 (from the simulator model)`
2. `People = MLD × 10⁶ ÷ 135 L · tankers a day = MLD × 10⁶ ÷ 12,000 L · pools a year = ML/yr ÷ 2.5`
3. `Peak demand = average × 1.3 for towers and adiabatic (April–May), × 1 otherwise`
4. `Shortfall: municipal = peak × supply cut · groundwater = peak × borewell failure share · treated wastewater = 0`
5. `Tanker trips a day = shortfall × 10⁶ ÷ 12,000 · monthly bill = trips × 30 × tanker price`
6. `Storage days = storage ML ÷ shortfall MLD`

WHAT EACH EQUATION DOES, AND WHY IT TAKES THIS FORM

- Daily water (MLD) comes straight from the simulator model: annual on-site (and optionally power-station) ML ÷ 365.
- 135 L is the CPHEEO norm for one person's daily urban water; a tanker carries 12,000 L; an Olympic pool holds 2.5 ML. These convert volumes into things people can picture.
- Evaporative designs peak in April–May at about 1.3× the average, exactly when supply is cut, so the drought test uses the peak.
- The shortfall depends on the source: a municipal supply cut, or the share of borewells that fail. Treated wastewater is assumed unaffected.

ASSUMED VALUES AND THE RATIONALE FOR EACH

VALUE	USED	WHY THIS VALUE (SOURCE)	BASIS
Tanker / pool	`12,000 L / 2.5 ML`	Standard Bengaluru tanker; Olympic pool volume. (Common units)	Convention

Peak factor	×1.3 for towers and adiabatic	April–May demand above the annual average. (Report 6)	Estimate
Borewell failure	47% (6,997 ÷ 14,781 public borewells, March 2024)	Observed failure share in Bengaluru's 2024 crisis. (BWSSB via press)	Reported

FURTHER NOTES FROM THE ATLAS

- 135 L/person/day (CPHEEO); Olympic pool 2.5 ML; BWSSB delivers ~1,935 MLD; Bangalore East sustainable groundwater ~45 MLD.
- 47% borewell failure = 6,997 ÷ 14,781 public borewells (March 2024). Tanker price is your input.

WORKED CALCULATION WITH THE SLIDER VALUES ABOVE

STEP	CALCULATION	RESULT
Water a day	(220.8) ML ÷ 365	**0.60 MLD**
People	$0.60 \times 10^6 \div 135$	**4,480**
Tankers a day	$0.60 \times 10^6 \div 12{,}000$	**50.4**
Peak demand	0.60 × 1.3	**0.79 MLD**
Shortfall	0.79 × 30% cut	**0.24 MLD**
Tanker bill	19.7 trips × 30 × ₹1,500	**₹8.8 lakh/month**

07 · Unit economics

07 · UNIT ECONOMICS

15. Levelised cost per IT-kWh

On the atlas panel: *Levelised cost per IT-kWh. LCOD = (capex × CRF + opex) ÷ IT energy delivered*

INPUTS TAKEN FROM THE SLIDERS AND SETTINGS

SETTING	VALUE USED
Capex (₹ cr/MW)	₹61.1
WACC	11.0%
Average IT draw	63%
PUE	1.50
Energy tariff (₹/kWh)	₹6.60
Other opex (₹ cr/MW/yr)	₹2.25
Case	A · colo

FORMULAS

1. CRF = $r(1 + r)^n \div ((1 + r)^n - 1)$, with n = 20 years
2. IT energy (MU per MW) = utilisation × 8.76
3. Capital charge (₹ cr/MW/yr) = capex × CRF
4. Power (₹ cr/MW/yr) = IT energy × PUE × tariff × 0.1 + demand charge
5. LCOD (₹/kWh) = (capital + power + other opex) ÷ IT energy × 10
6. Break-even rent (₹/kW-month) = annual cost × 10^7 ÷ 12,000

WHAT EACH EQUATION DOES, AND WHY IT TAKES THIS FORM

- A data centre is paid for up front but earns over ~20 years. The capital recovery factor CRF = $r(1+r)^n \div ((1+r)^n - 1)$ turns capex into the equal yearly payment that repays it with interest at rate r: the same maths as a home loan EMI.
- IT energy per MW = utilisation × 8.76 MU. Electricity = IT energy × PUE × tariff, plus a fixed demand charge per MW.
- Levelised cost = (capital charge + electricity + other opex) ÷ IT energy. Dividing by IT energy (not facility energy) gives the cost of the useful output.
- ×0.1 converts MU × ₹/kWh to ₹ crore; ×10 converts ₹ crore per MU to ₹ per kWh; ÷ 12,000 spreads a yearly ₹ crore per MW over 1,000 kW × 12 months.

ASSUMED VALUES AND THE RATIONALE FOR EACH

VALUE	USED	WHY THIS VALUE (SOURCE)	BASIS
Life	20 years	Building and core plant life used for levelising. (Report 7)	Series assumption
WACC default	11%	Indian infrastructure cost of capital. (Report 7)	Series assumption
Demand charge	₹0.99 / 0.66 / 0.66 cr per MW-yr by preset	State demand charges on contract demand for each case. (Report 7)	Estimate
Other opex	O&M, staff, insurance, SG&A per MW	Bottom-up opex by case. (Report 7)	Estimate
Presets	Case A colo Mumbai, B hyperscale, C AI	Report 7 worked cases. (Report 7)	Series assumption

FURTHER NOTES FROM THE ATLAS

- Demand charges fixed per preset: ₹0.99, 0.66 and 0.66 crore per MW a year.
- Capex includes land; other opex = O&M, staff, insurance, SG&A per MW (Report 7).
- ×0.1 converts MU × ₹/kWh into ₹ crore; ×10 converts ₹ crore per MU into ₹ per kWh.

WORKED CALCULATION WITH THE SLIDER VALUES ABOVE

STEP	CALCULATION	RESULT
CRF	$11.0\% \times (1+11.0\%)^{20} \div ((1+11.0\%)^{20} - 1)$	**0.1256**
Capital charge	₹61.1 cr × 0.1256	**₹7.67 cr/MW/yr**
IT energy	63% × 8.76	**5.52 MU/MW**
Electricity	5.52 × 1.50 × ₹6.60 × 0.1 + ₹0.99 demand	**₹6.45 cr/MW/yr**
Total cost	7.67 + 6.45 + 2.25 other	**₹16.38 cr/MW/yr**
LCOD	16.38 ÷ 5.52 × 10	**₹29.7/IT-kWh**
Break-even rent	$16.38 \times 10^7 \div 12{,}000$	**₹13,647/kW-month**

07 · UNIT ECONOMICS

16. GPU-hour cost

On the atlas panel: *GPU-hour cost in India. GPU + network capex, colo hosting at ₹7,000/kW/month, power at 80% draw, ₹70k/yr ops. Dashed lines are market prices.*

INPUTS TAKEN FROM THE SLIDERS AND SETTINGS

SETTING	VALUE USED
All-in capex per GPU (₹ lakh)	₹36.5
Economic life (years)	5 yrs
WACC	11.0%
Tariff (₹/kWh)	₹7.0
Your utilisation	80%
GPU	H100 (HGX, ~US$275k per 8)

FORMULAS

1. Fixed (₹/yr) = capex (₹ lakh) × 10^5 × CRF(WACC, life) + kW × ₹7,000 × 12 + ₹70,000
2. Power (₹/hr) = kW per GPU × 0.8 load × PUE × tariff
3. Cost (₹/GPU-hr) = fixed ÷ (8,760 × utilisation) + power
4. Break-even utilisation = fixed ÷ (8,760 × (market price − power))

WHAT EACH EQUATION DOES, AND WHY IT TAKES THIS FORM

- A GPU is written off over 4–6 years (it is obsolete quickly), so its capex is annualised with a short-life CRF.
- Hosting is rented by the kW: kW per GPU × ₹7,000 per kW-month × 12. Support and software add ~₹70,000 a year.
- Fixed cost is spread over the hours the GPU is actually busy (8,760 × utilisation); electricity is paid per hour of use (kW × 0.8 average draw × PUE × tariff).
- Break-even utilisation solves fixed ÷ (8,760 × u) + power = market price for u.

ASSUMED VALUES AND THE RATIONALE FOR EACH

VALUE	USED	WHY THIS VALUE (SOURCE)	BASIS

kW per GPU	H100/H200 1.4, B200 2.0 (incl. server share)	GPU TDP plus CPU, memory, network share per GPU. (Vendor specs; Report 7)	Estimate
PUE	1.35 (1.30 for B200)	IndiaAI tender ceiling; newer liquid-cooled systems run lower. (MeitY tender; Report 7)	Estimate
Hosting	₹7,000 per kW-month	Indian colo rate for high-density space, power excluded. (Report 7 market survey)	Estimate
Support	₹70,000 per GPU-year	Software, support contracts and spares. (Series)	Series assumption
Load	0.8 average draw	GPUs under training load average ~80% of rated power (LBNL measured ~78%). (LBNL 2024)	Reported
Market prices	IndiaAI L1 ₹150, E2E ₹255.55, Yotta ₹356 per GPU-hour	Published prices used as comparison lines. (Providers, Oct 2026)	Reported

FURTHER NOTES FROM THE ATLAS

- Capex per GPU includes its share of the server, network and storage.
- kW per GPU incl. server share: H100/H200 1.4, B200 2.0; PUE 1.35 (1.30 for B200).
- Colo hosting ₹7,000 per kW-month; ₹70,000 a year for support, software and spares (series assumptions).
- Market prices: IndiaAI L1 ₹150, E2E ₹255.55, Yotta on-demand ₹356 per GPU-hour.

WORKED CALCULATION WITH THE SLIDER VALUES ABOVE

STEP	CALCULATION	RESULT
CRF	r = 11.0%, n = 5 years	**0.2706**
Fixed cost a year	₹36.5 lakh × 10^5 × 0.2706 + 1.4 × 7,000 × 12 + 70,000	**₹11,75,182**
Power per hour	1.4 × 0.8 × 1.35 × ₹7.0	**₹10.6**
Cost per GPU-hour	11,75,182 ÷ (8,760 × 80%) + 10.6	**₹178**
Break-even at ₹150	11,75,182 ÷ (8,760 × (150 − 10.6))	**96%**

08 · Green data centres

08 · GREEN DATA CENTRES

17. Green upgrade calculator

On the atlas panel: *Green upgrade calculator. Baseline: grid power, evaporative towers. Green: lower PUE, hourly renewables, dry or treated-water cooling.*

INPUTS TAKEN FROM THE SLIDERS AND SETTINGS

SETTING	VALUE USED
IT capacity (MW)	20 MW
Average IT load	70%
Baseline PUE	1.50
Green PUE	1.25
Hourly renewable share	80%
Grid tariff (₹/kWh)	₹8.00
Renewable landed cost (₹/kWh)	₹7.25
Green capex premium (₹ cr)	₹60

FORMULAS

1. IT energy (MU) = MW × utilisation × 8.76
2. Before = IT × PUE_0 · After = IT × PUE_1
3. Efficiency saving (₹ cr) = (before − after) × tariff × 0.1
4. Renewables saving (₹ cr) = after × RE share × (tariff − RE price) × 0.1
5. Freshwater saved (ML) = IT MU × 1.6 L/kWh · water saving (₹ cr) = ML × 0.009
6. CO_2 before = before × 710 t/MU · after = after × (1 − RE share) × 710
7. Abatement cost (₹/t) = (capex × CRF(10%, 15) − all savings) ÷ t abated
8. Payback (years) = capex ÷ (efficiency + water savings)

WHAT EACH EQUATION DOES, AND WHY IT TAKES THIS FORM

- Every saving is avoided quantity × its price: kWh saved × tariff, kWh switched to renewables × the price gap, litres not evaporated × the price of water.
- Removing cooling towers saves the tower's ~1.6 L per kWh of IT; water is valued at ₹90/kL, i.e. ₹0.009 crore per ML.
- CO_2 uses the CEA grid factor, 710 t per MU (0.71 t/MWh); renewable energy is counted at zero.
- Abatement cost = (annualised upgrade capex − all savings) ÷ tonnes avoided. Negative means the upgrade saves money while cutting carbon. Payback = capex ÷ yearly efficiency and water savings.

ASSUMED VALUES AND THE RATIONALE FOR EACH

VALUE	USED	WHY THIS VALUE (SOURCE)	BASIS
Baseline WUE	1.6 L/kWh	Evaporative tower plant removed by the upgrade. (Report 8)	Estimate
Water value	₹90/kL	Indicative cost of industrial water in Bengaluru. (Report 8)	Estimate
CO_2 factor	710 t/MU	CEA grid factor (0.71 t/MWh). (CEA)	Reported
Upgrade finance	CRF(10%, 15 years)	Retrofit equipment life and cost of capital. (Report 8)	Series assumption

FURTHER NOTES FROM THE ATLAS

- Baseline: grid power, evaporative towers at WUE 1.6; upgrade removes evaporative cooling.
- Water valued at ₹90/kL; grid factor 0.710 t/MWh (CEA).
- Karnataka pays ₹20 crore one-time for PUE ≤ 1.20 (shown, not added to savings).

WORKED CALCULATION WITH THE SLIDER VALUES ABOVE

STEP	CALCULATION	RESULT
IT energy	20 × 70% × 8.76	**122.6 MU**
Before / after	122.6 × 1.50 / × 1.25	**184.0 / 153.3 MU**
Efficiency saving	(184.0 − 153.3) × ₹8.00 × 0.1	**₹24.5 cr**
Renewables saving	153.3 × 80% × (8.00 − 7.25) × 0.1	**₹9.2 cr**
Water saving	122.6 × 1.6 = 196 ML × 0.009	**₹1.8 cr**
CO_2 avoided	184.0 × 710 − 153.3 × 0.20 × 710	**1,08,843 t**
Abatement cost	(₹60 cr × CRF(10%,15) − savings) × 10^7 ÷ 1,08,843	**₹-2,536/t**
Payback	60 ÷ (24.5 + 1.8)	**2.3 yrs**

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08 · GREEN DATA CENTRES

18. Maturity self-check

On the atlas panel: *Maturity self-check. Pick a level for each dimension (India-calibrated). The radar compares you with today's Indian leaders.*

INPUTS TAKEN FROM THE SLIDERS AND SETTINGS

SETTING	VALUE USED
Energy level	L3
Carbon level	L3
Water level	L2
Materials level	L2
Circularity level	L2
Heat level	L1
Backup level	L2
Transparency level	L2

FORMULAS

1. Average level = sum of the 8 levels ÷ 8
2. Weakest dimension = the lowest level

WHAT EACH EQUATION DOES, AND WHY IT TAKES THIS FORM

- The average shows overall maturity; the weakest dimension matters more in practice, because a green claim fails at its weakest link (e.g. excellent PUE but unknown water use).

ASSUMED VALUES AND THE RATIONALE FOR EACH

VALUE	USED	WHY THIS VALUE (SOURCE)	BASIS
Levels	L1–L5 per dimension	India-calibrated maturity ladder; "leader today" is a series judgement. (Report 8)	Series assumption

FURTHER NOTES FROM THE ATLAS

- Level descriptions are the Report 8 India-calibrated ladder. "Indian leader today" is a series judgment.

WORKED CALCULATION WITH THE SLIDER VALUES ABOVE

STEP	CALCULATION	RESULT
Levels	3 + 3 + 2 + 2 + 2 + 1 + 2 + 2	**= 17**
Average	17 ÷ 8	**2.13**
Weakest	Heat	**L1**

09 · State policies

09 · STATE POLICIES

19. India map

On the atlas panel: *India at a glance. Colour the map by a measure, then tap or hover a state for details.*

INPUTS TAKEN FROM THE SLIDERS AND SETTINGS

SETTING	VALUE USED
Map measure	`Policy index`

FORMULAS

1. `Policy index = Σ (score × weight) ÷ Σ weights (uses your chapter-09 weights)`
2. `Colour intensity t = (value − min) ÷ (max − min); for MW, t = log₁₀(value + 1) ÷ log₁₀(max + 1)`
3. `Sustainability rank is inverted so darker = better`

WHAT EACH EQUATION DOES, AND WHY IT TAKES THIS FORM

- The map's policy index is the same weighted average as the attractiveness index, using your chapter 9 weights.
- Colour intensity rescales values to 0–1 so the lightest state is the minimum and the darkest the maximum. Megawatts span 0 to ~900, so a log scale stops Mumbai washing out every other state.

ASSUMED VALUES AND THE RATIONALE FOR EACH

VALUE	USED	WHY THIS VALUE (SOURCE)	BASIS
Colour scale	`min–max; log for MW`	Makes small and large values visible on one map. (Display)	Convention

FURTHER NOTES FROM THE ATLAS

- Live and pipeline MW: Cushman & Wakefield H1 2026. NCR is one market, shown on UP, Haryana and Delhi together.
- Announced gigawatt campuses (Vizag, Jamnagar) are outside the pipeline count.
- Boundaries: official map of India (udit-001/india-maps-data), simplified.

WORKED CALCULATION WITH THE SLIDER VALUES ABOVE

STEP	CALCULATION	RESULT
State	`Maharashtra`	
Power	`8 × 25`	**200**
Renewables	`6 × 15`	**90**
Connectivity	`10 × 15`	**150**
Incentives	`8 × 15`	**120**
Land	`6 × 10`	**60**
Water	`6 × 10`	**60**
Regulatory	`8 × 10`	**80**
Index	`760 ÷ 100`	**7.60 / 10**

09 · STATE POLICIES

20. Attractiveness index

On the atlas panel: *Attractiveness index. States scored 0–10 per factor (Report 9's judgment). Change how much each factor matters.*

INPUTS TAKEN FROM THE SLIDERS AND SETTINGS

SETTING	VALUE USED
Power	25
Renewables	15
Connectivity	15
Incentives	15
Land	10
Water	10
Regulatory	10
Weight presets	Report weights

FORMULAS

1. Index = Σ (score × weight) ÷ Σ weights

WHAT EACH EQUATION DOES, AND WHY IT TAKES THIS FORM

- Each factor is scored 0–10 (Report 9's judgement). Multiplying by its weight and dividing by the sum of weights gives a weighted average on the same 0–10 scale, whatever weights you choose.
- Weights express priorities: an AI campus cares about power and renewables, a colo about connectivity. Changing them re-ranks states without changing any score.

ASSUMED VALUES AND THE RATIONALE FOR EACH

VALUE	USED	WHY THIS VALUE (SOURCE)	BASIS
Factor scores	0–10 per state	Report 9 judgement from policy documents, power, connectivity and water data. (Report 9)	Series assumption
Default weights	power 25, renewables 15, connectivity 15, incentives 15, land 10, water 10, regulatory 10	Reflect what drives site choice for a typical project. (Report 9)	Series assumption

FURTHER NOTES FROM THE ATLAS

- Seven factor scores 0–10 per state are the Report 9 judgment, not published data.
- Report weights: power 25, renewables 15, connectivity 15, incentives 15, land 10, water 10, regulatory 10.

WORKED CALCULATION WITH THE SLIDER VALUES ABOVE

STEP	CALCULATION	RESULT
State	Maharashtra	
Power	8 × 25	**200**
Renewables	6 × 15	**90**
Connectivity	10 × 15	**150**
Incentives	8 × 15	**120**
Land	6 × 10	**60**
Water	6 × 10	**60**
Regulatory	8 × 10	**80**
Index	760 ÷ 100	**7.60 / 10**

10 · Where to build

10 · WHERE TO BUILD

21. City ranking (Steps A–E)

On the atlas panel: *Ranking engine. Choose a use case or set your own weights. The filters, scores, map and table all update live.*

INPUTS TAKEN FROM THE SLIDERS AND SETTINGS

SETTING	VALUE USED
Power	25
Connectivity	15
Latency	10
Water	15
Climate	5
Hazards	10
Land	5
Policy	10
Demand & talent	5
Hard filter: 220 kV+ substation within	25 km
Exclude seismic zone V	on
Allow over-exploited groundwater with a treated-water contract	on
Exclude very-high cyclone or flood sites	off
Use case	Balanced (default)
Method	Weighted sum Σ
Normalisation	Min–max

FORMULAS

1. A · Hard filters: keep a city only if seismic zone ≤ IV, nearest 220 kV+ substation ≤ your distance limit, not very-high cyclone or flood (if that filter is on), and groundwater not over-exploited unless treated water is assumed
2. Latency (ms) = mean over Mumbai, Delhi, Chennai, Bengaluru of 2 × great-circle km × 1.4 ÷ 200 = 0.014 × km
3. B · Min–max: $s = 10 \times (x - \min) \div (\max - \min)$ over the cities that pass the filters; latency flipped: $s = 10 \times (\max - x) \div (\max - \min)$
4. B · Report rubric (alternative): factor scores as published in Report 10; latency = 10 × (26 − RTT) ÷ 16, clipped 0–10
5. C · Weighted sum: score = $\Sigma\, w_i \times s_i$ with $\Sigma\, w_i = 1$ (sliders are rescaled to sum to 1)
6. C · Geometric: score = $\Pi \max(s_i, 0.5)^{w_i}$ (floor 0.5 so a zero does not wipe out the city)
7. Tiers: A = top 30% of cities that pass, B = next 40%, C = rest; excluded cities shown separately
8. E · Monte Carlo: 1,000 runs; each weight × U(0.8, 1.2) then rescaled; each factor score × U(0.8, 1.2), clipped 0–10; count how often each city is top 3 in its state and top 10 nationally

WHAT EACH EQUATION DOES, AND WHY IT TAKES THIS FORM

- Hard filters come first because no score can compensate for a deal-breaker (no 220 kV substation nearby, seismic zone V, over-exploited groundwater without treated water).
- Latency is estimated from distance: signals in fibre travel ~200 km per millisecond, routes are ~1.4× straight-line, and a round trip doubles it, so RTT ≈ 0.014 ms per km.

- Min–max scaling puts every factor on 0–10 across the cities that passed, so factors with different units (₹, km, ms) can be added. Latency is flipped because lower is better.
- The weighted average then ranks cities; weights are your priorities. The Monte Carlo varies weights and scores ±20% 1,000 times to test whether a city's rank depends on fine-tuning.

ASSUMED VALUES AND THE RATIONALE FOR EACH

VALUE	USED	WHY THIS VALUE (SOURCE)	BASIS
Fibre speed and route factor	`200 km/ms; 1.4× straight line`	Light in glass travels ~2/3 of c; cables follow roads and coasts. (Physics; network practice)	Physics
Hard-filter limits	`zone ≤ IV; substation within your km; cyclone/flood optional`	Deal-breakers that no score should offset. (Report 10)	Series assumption
Monte Carlo	`1,000 runs, ±20% on weights and scores`	Tests whether rank depends on fine-tuning. (Report 10)	Series assumption

FURTHER NOTES FROM THE ATLAS

- Power score = 50% tariff + 50% grid and renewables judgement (Report 10). Connectivity, water, climate, hazards, land, policy and demand are Report 10 rubric scores (0–10).
- Substation names and distances are estimates; spare capacity is unpublished everywhere.
- Effective tariff = energy charge + demand charge ÷ 584 h (80% load factor).
- Seismic zones follow IS 1893:2016 (the 2025 revision was withdrawn in March 2026).
- The Monte Carlo uses a fixed seed, so results repeat until you press "Re-run".

WORKED CALCULATION WITH THE SLIDER VALUES ABOVE

STEP	CALCULATION	RESULT
Cities passing filters	`of 50`	**48**
Top city	`Hyderabad periphery`	**7.27**
Latency	`mean RTT to 4 metros = 0.014 × km`	**10.0 ms**
Power	`6.2 × weight 0.250`	**1.54**
Connectivity	`6.3 × weight 0.150`	**0.94**
Latency	`9.8 × weight 0.100`	**0.98**
Water	`8.3 × weight 0.150`	**1.25**
Climate	`5.7 × weight 0.050`	**0.29**
Hazards	`10.0 × weight 0.100`	**1.00**
Land	`6.7 × weight 0.050`	**0.33**
Policy	`5.0 × weight 0.100`	**0.50**
Demand & talent	`8.9 × weight 0.050`	**0.44**
Score	`sum of the above (or geometric mean, if chosen)`	**7.27**

11 · Forecast to 2035

11 · FORECAST TO 2035

22. Forecast to 2035

On the atlas panel: *Forecast engine. Four headline sliders explain most of the uncertainty: the S-curve ceiling, how much of the planned pipeline gets built, utilisation and the grid's carbon intensity. Open "More assumptions" for the rest.*

INPUTS TAKEN FROM THE SLIDERS AND SETTINGS

SETTING	VALUE USED
Show values for	2030
S-curve ceiling K	25.0 GW
Planned pipeline actually built	60%
Utilisation in 2035	65%
Grid CO_2 factor in 2035	0.535 t/MWh
S-curve growth rate r	0.26
Growth rate g (CAGR model)	25%
Under-construction delivered	95%
Extra GW campuses outside 7 markets by 2030	0 MW
Delivery factor after 2030	0.90
Utilisation in 2026	55%
PUE in 2035	1.35
Clean power share in 2035	60%
Liquid-cooling imitation q	0.40
Liquid / dry ceiling m	85%
WUE evaporative	1.80 L/kWh
Learning exponent b	0.10 (LR 7%)
Equipment inflation π	4.0%/yr
AI facility premium a	+50%
GPU price decline d	20%/yr
H100-class price 2026	₹220
India demand growth	6.41%/yr
City allocation exponent k	8
Incumbency weight α	0.50
Capacity model	Integrated (base)
Scenario	Base

FORMULAS

1. CAGR: $C(y) = 1{,}611 \times (1 + g)^{(y - 2025)}$
2. Logistic: $C(y) = K \div (1 + e^{(-r(y - t_0))})$, with $t_0 = 2025 + \ln(K \div 1{,}611 - 1) \div r$ so the curve passes through end-2025

3. Pipeline: C(y) = 1,789 + 509 × c_UC × f_UC(y) + 3,351 × c_P × f_P(y) + X × f_P(y); f_UC = 0.4, 0.9, 1 for 2026, 2027, 2028+; f_P = 0, 0.12, 0.37, 0.65, 1 for 2026–2030
4. Integrated: 2026–2030 = mean(pipeline, logistic); 2031–2035 = logistic × delivery factor (base 0.9)
5. Electricity: IT TWh = MW × u × 8.76 ÷ 1,000; facility TWh = IT × PUE; u and PUE move linearly 2026 → 2035
6. India requirement (BU) = 1,694 × (1 + g_grid)^(y − 2024); share = facility TWh ÷ BU
7. Carbon: net Mt = facility TWh × EF(y) × (1 − clean(y)); location-based Mt = facility TWh × EF(y). EF and clean share are piecewise linear through 2026, 2030 and 2035
8. Liquid / dry share (Bass): L = 0.20 + (m − 0.20) × F(y − 2025), F(t) = (1 − e^(−(p+q)t)) ÷ (1 + (q/p)e^(−(p+q)t))
9. WUE = L × 0.1 + (1 − L) × (e × WUE_evap + (1 − e) × 0.5), where evaporative share e of the rest falls 0.69 → 0.50; MLD = IT TWh × WUE × 2.7397
10. Build cost (air) = 58.4 × (C(y) ÷ 1,611)^(−b) × (1 + π)^(y − 2025); AI = air × (1 + a); blended = air × (1 + a × AI share), AI share 0.4 → 0.6 (2030) → 0.7 (2035)
11. Capex (₹ crore) = Σ ΔMW × blended cost; excludes land and IT/GPU hardware
12. GPU price (₹/hr) = P_0 × (1 − d)^(y − 2026)
13. Cities 2026–2030: increment over 1,789 MW split 8.6% Vizag, 5.2% other, rest by C&W pipeline shares; 2031–2035 each year's increment split α × current share + (1 − α) × score^k ÷ Σ score^k
14. Monte Carlo: 1,000 runs; K lognormal (σ 0.36) around your value, other inputs triangular or uniform around your values (ranges in the parameter table); K and utilisation correlated (ρ ≈ 0.5). Band = 10th–90th percentile

WHAT EACH EQUATION DOES, AND WHY IT TAKES THIS FORM

- Three ways to project capacity: constant growth (CAGR), an S-curve that slows as the market matures (logistic, ceiling K), and a bottom-up pipeline of projects under construction and planned, each discounted by how much usually gets built.
- The logistic curve is anchored so it passes through the actual end-2025 capacity (1,611 MW); r sets how fast it rises.
- The integrated forecast averages pipeline and logistic to 2030 (projects are known), then follows the S-curve, discounted by a delivery factor, because pipelines beyond five years are speculative.
- Electricity = MW × utilisation × 8.76 ÷ 1,000 (TWh of IT) × PUE.

ASSUMED VALUES AND THE RATIONALE FOR EACH

VALUE	USED	WHY THIS VALUE (SOURCE)	BASIS
Base	1,611 MW end-2025; 1,789 MW operational H1 2026	Cushman & Wakefield capacity data. (C&W)	Reported
Pipeline	509 MW under construction; 3,351 MW planned	C&W pipeline to 2030. (C&W)	Reported
Phasing	UC 40/90/100% 2026–28; planned 0/12/37/65/100% 2026–30	Typical build durations. (Report 11)	Estimate
Ceiling K default	25 GW	Middle of published long-run estimates. (Report 11)	Series assumption
Delivery factor	0.9 after 2030	Backtests show forecasts overshoot. (Report 11 backtest)	Estimate

FURTHER NOTES FROM THE ATLAS

- History is Report 11's harmonised series (375 MW in 2019 to 1,611 MW at end-2025); the C&W H1 2026 figure of 1,789 MW is the anchor.
- K (the ceiling) is a judgement, not a statistical fit: seven early-growth data points cannot identify it.

- State electricity denominators are estimates (≈6%/yr from FY25), not CEA state tables.
- City scores default to Report 11's illustrative values; switch to the live chapter-10 scores below.

WORKED CALCULATION WITH THE SLIDER VALUES ABOVE

STEP	CALCULATION	RESULT
Year	selected	**2030**
CAGR	1,611 × (1 + 0.25)^(2030 − 2025)	**4,916 MW**
Logistic anchor t_0	2025 + ln(25,000 ÷ 1,611 − 1) ÷ 0.26	**2035.3**
Logistic	25,000 ÷ (1 + e^(−0.26 × (2030 − 2035.3)))	**5,044 MW**
Pipeline	1,789 + 509 × 0.95 × 1 + 3,351 × 0.60 × 1	**4,283 MW**
Integrated	mean(4,283, 5,044)	**4,663 MW**

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23. Power vs water

On the atlas panel: *Power versus water: every cooling technology. Each box spans the typical annual PUE and on-site water (WUE) range in Indian climates. Dots are published fleet figures. Switch on the grid's water to see where each design*

INPUTS TAKEN FROM THE SLIDERS AND SETTINGS

SETTING	VALUE USED
Grid water intensity	`3.0 L/kWh`
Water counted	`On-site only`

FORMULAS

1. `On-site view: box = [WUE low, WUE high] × [PUE low, PUE high] from the Report 12 cooling table`
2. `Total-water view: source WUE = WUE + PUE × grid water intensity (EWIF); box corners move by PUE × EWIF`
3. `Grid water intensity default 3.0 L/kWh at the meter; Karnataka ~1.8, Maharashtra ~3.1`

WHAT EACH EQUATION DOES, AND WHY IT TAKES THIS FORM

- On-site WUE alone favours waterless cooling. But every kWh the site buys also evaporates water at a coal plant, so total water per kWh of IT = on-site WUE + PUE × grid water intensity.
- That moves every box right by PUE × grid water: designs with a high PUE are pushed furthest, which is why a "waterless" air-chiller site can end up thirstier than an efficient adiabatic one.

ASSUMED VALUES AND THE RATIONALE FOR EACH

VALUE	USED	WHY THIS VALUE (SOURCE)	BASIS
Grid water	`3.0 L/kWh at the meter (Karnataka ~1.8)`	CEA FY25 generation mix × water per source, ÷ (1 − 17% losses). (Report 12)	Estimate
Technology ranges	`e.g. towers WUE 1.5–2.5, PUE 1.35–1.60`	Engineering ranges for Indian climates cross-checked with LBNL and operators. (Report 12)	Estimate

FURTHER NOTES FROM THE ATLAS

- Ranges are engineering estimates for Indian climates [proxy], cross-checked against LBNL 2024 and operator data.
- Fleet dots: Microsoft FY25 (global and Asia-Pacific), NTT DATA FY24 portfolio, LBNL US average 2023, Google PUE 2025 with fleet WUE 2024 (different years). Bases differ (withdrawal vs consumption).

WORKED CALCULATION WITH THE SLIDER VALUES ABOVE

STEP	CALCULATION	RESULT
Cooling towers	`1.5–2.5 + (1.35–1.60) × 3.0`	**`5.6–7.3 L/kWh`**
Adiabatic / hybrid	`0.2–0.8 + (1.20–1.40) × 3.0`	**`3.8–5.0`**
Air-cooled chillers	`0–0.05 + (1.45–1.75) × 3.0`	**`4.3–5.3`**
Liquid + dry coolers	`0–0.05 + (1.10–1.25) × 3.0`	**`3.3–3.8`**

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24. Hourly profile

On the atlas panel: *One hot day, hour by hour. A 50 MW hyperscale site in Hyderabad on a May design day, 42 MW of IT load all day. Bars are water drawn each hour; the line is the second measure you choose.*

INPUTS TAKEN FROM THE SLIDERS AND SETTINGS

SETTING	VALUE USED
From	10:00
To (end of hour)	18:00
"Dry at noon": switch off evaporation in a window	off
Cooling design	Adiabatic
Line	Facility power (MW)

FORMULAS

1. Temperature: Report 12 May design day, 27 °C at 05:00 to 40 °C at 15:00 (2-hourly points, interpolated)
2. Adiabatic: PUE(T) = 1.22 + 0.015 × max(0, T − 25); WUE(T) = 0 below 26 °C, else min(1.6, 0.1 + 0.107 × (T − 26))
3. Cooling towers: WUE(T) = 1.9 + 0.4 × (T − 27) ÷ 13; PUE(T) = 1.43 + 0.15 × (T − 27) ÷ 13
4. Dry + liquid: WUE = 0.03; PUE(T) = 1.15 + 0.008 × max(0, T − 25)
5. Water (kL/h) = IT MW × WUE; facility MW = IT MW × PUE
6. Dry at noon: inside the window WUE = 0 and PUE + penalty; extra cost = extra MWh × exchange price

WHAT EACH EQUATION DOES, AND WHY IT TAKES THIS FORM

- Adiabatic coolers run dry until the air is hot (~26 °C), then wet their pads; water use rises with temperature and is capped at ~1.6 L/kWh. PUE rises gently with heat because fans and trim chillers work harder.
- Water each hour = IT MW × WUE at that hour's temperature (MW × L/kWh = kL/h). Summing 24 hours gives the day; the peak hour sizes pipes and storage.
- "Dry at noon" switches pads off in a window: water falls to zero there but PUE rises, and the extra MWh are priced at that hour's exchange price.

ASSUMED VALUES AND THE RATIONALE FOR EACH

VALUE	USED	WHY THIS VALUE (SOURCE)	BASIS
Design day	27 °C at 05:00 to 40 °C at 15:00 (Hyderabad, May)	A hot pre-monsoon day. (Report 12)	Estimate
Adiabatic curves	PUE 1.22 + 0.015(T − 25); WUE 0.1 + 0.107(T − 26), max 1.6	Fitted so the day totals 905 kL and 1,358 MWh, as in Report 12. (Report 12)	Estimate
IT load	42 MW all day	50 MW site at 84% load. (Report 12)	Series assumption

FURTHER NOTES FROM THE ATLAS

- Adiabatic functions and design day are Report 12 models; they give 905 kL and 1,358 MWh for the day.
- Tower and dry curves are illustrative [model], tuned to the Report 12 ranges (towers nearly flat at 1.9–2.3 L/kWh; dry ≈ 0).
- Exchange prices are shaped to June 2026 IEX hourly averages (trough ₹1.17 at 11:00, peak ₹9.90 at 22:30) [reported, shape is a proxy].
- The +0.10 PUE "dry" penalty is an assumption for running chillers with pads off.
- Report 12 says an 11:00–16:00 window halves the peak hour; on its own profile the 16:00–17:00 hours stay hot, so a window to about 18:00 is needed to cut the peak substantially.

WORKED CALCULATION WITH THE SLIDER VALUES ABOVE

STEP	CALCULATION	RESULT

27 °C	PUE 1.22 + 0.015 × 2 · WUE 0.1 + 0.107 × 1	**1.25 · 0.21 L/kWh → 8.7 kL/h at 42 MW**
32 °C	PUE 1.22 + 0.015 × 7 · WUE 0.1 + 0.107 × 6	**1.32 · 0.74 L/kWh → 31.2 kL/h at 42 MW**
36 °C	PUE 1.22 + 0.015 × 11 · WUE 0.1 + 0.107 × 10	**1.39 · 1.17 L/kWh → 49.1 kL/h at 42 MW**
40 °C	PUE 1.22 + 0.015 × 15 · WUE 0.1 + 0.107 × 14	**1.44 · 1.60 L/kWh → 67.1 kL/h at 42 MW**

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25. Annual hours

On the atlas panel: *All 8,760 hours of a year. The same adiabatic site, run through a whole year of typical temperatures. The histogram counts how many hours fall at each rate of water draw; the grid shows when they happen.*

INPUTS TAKEN FROM THE SLIDERS AND SETTINGS

SETTING	VALUE USED
City	Hyderabad
Measure	Water (kL/h)

FORMULAS

1. Hourly temperature = monthly mean minimum + (mean maximum − mean minimum) × shape(h); shape rises as a half-cosine from 05:00 to 15:00 and falls over the next 14 h
2. Water each hour (kL) = IT MW × WUE(T), with the adiabatic WUE(T) above
3. Annual WUE = Σ water ÷ Σ IT energy; wet-mode hours = hours above 28 °C
4. Peak-month factor = highest month's daily water ÷ annual daily average

WHAT EACH EQUATION DOES, AND WHY IT TAKES THIS FORM

- A year has 8,760 hours with very different temperatures; the same hourly WUE(T) rule is applied to each and summed, so annual WUE = total water ÷ total IT energy.
- Hours above 28 °C are counted as "wet" because that is when the pads run; the peak month divided by the average shows how much bigger the April–May supply must be.

ASSUMED VALUES AND THE RATIONALE FOR EACH

VALUE	USED	WHY THIS VALUE (SOURCE)	BASIS
Climate	approximate IMD monthly normals	Smooth normals miss heatwaves; replace with typical-year files for design. (IMD (approx.))	Estimate

FURTHER NOTES FROM THE ATLAS

- Monthly mean maxima and minima are approximate IMD normals [proxy]; replace with ISHRAE typical-year files.
- Smooth normals miss heatwaves, so true peaks are higher; Report 12 uses 0.45 L/kWh for Hyderabad to allow for this.
- The model uses dry-bulb only. In humid Mumbai and Chennai, evaporation cools less, so real adiabatic designs save less water than shown.

WORKED CALCULATION WITH THE SLIDER VALUES ABOVE

STEP	CALCULATION	RESULT
24 °C night	below 26 °C: pads off	**WUE 0**
30 °C morning	0.1 + 0.107 × 4	**0.53 L/kWh**
38 °C afternoon	0.1 + 0.107 × 12	**1.38 L/kWh**

Annual WUE	Σ (IT MWh × WUE each hour) ÷ Σ IT MWh over 8,760 hours	**shown in the chart readout**

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26. Tower water balance

On the atlas panel: *Inside a cooling tower. Where each litre of make-up water goes. Raising cycles of concentration (cleaner water, softening or RO) is the cheapest saving at existing tower sites.*

INPUTS TAKEN FROM THE SLIDERS AND SETTINGS

SETTING	VALUE USED
Cycles of concentration (CoC)	4.0
Share of heat removed by evaporation	85%

FORMULAS

1. Heat rejected Q = 1.25 kWh of heat per kWh IT (chiller COP 5–7)
2. Evaporation E = Q × latent share × 3,600 ÷ 2,430 kJ/kg
3. Blowdown B = E ÷ (CoC − 1)
4. Drift D = 0.002 L per kWh of heat
5. WUE = E + B + D per kWh IT

WHAT EACH EQUATION DOES, AND WHY IT TAKES THIS FORM

- Heat to reject is ~1.25 kWh per kWh of IT (IT heat plus chiller work). A share of it (the latent share) leaves as evaporation: water = heat × 3,600 kJ/kWh ÷ 2,430 kJ/kg.
- Blowdown keeps salts in check: B = E ÷ (CoC − 1). Drift is tiny spray loss. Make-up WUE = E + B + D.

ASSUMED VALUES AND THE RATIONALE FOR EACH

VALUE	USED	WHY THIS VALUE (SOURCE)	BASIS
Heat rejected	1.25 kWh per kWh IT	IT heat plus chiller work at COP 5–7. (Physics)	Physics
Latent share	0.75–0.95 (default 0.85)	Share of tower heat removed by evaporation. (Report 12)	Estimate
Latent heat	2,430 kJ/kg	At ~30 °C water temperature. (Physics)	Physics

FURTHER NOTES FROM THE ATLAS

- Latent heat at ~30 °C; latent share 0.75–0.95 (default 0.85) [proxy].
- NTT DATA's evaporative sites reported 1.42–2.93 L/kWh in FY22–24 [reported].

WORKED CALCULATION WITH THE SLIDER VALUES ABOVE

STEP	CALCULATION	RESULT
Evaporation	1.25 × 0.85 × 3,600 ÷ 2,430	**1.57 L/kWh**
Blowdown	1.57 ÷ (4.0 − 1)	**0.52 L/kWh**
Drift	0.002 × 1.25	**0.0025**
Tower WUE	1.57 + 0.52 + 0.0025	**2.10 L/kWh**

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27. Site calculator

On the atlas panel: *Your site: water, power and peaks. Report 12's calculator defaults. Pick a design and a grid; every output updates.*

INPUTS TAKEN FROM THE SLIDERS AND SETTINGS

SETTING	VALUE USED
IT capacity	50 MW
Average utilisation	80%
Round-the-clock renewable share	0%
Landed power cost	₹7.5/kWh
Water storage autonomy	48 h
Cooling design	Adiabatic / hybrid
Grid (water intensity)	India average (3.0 L/kWh)

FORMULAS

1. IT energy (MWh/yr) = IT MW × utilisation × 8,760; facility energy = IT energy × PUE
2. Direct water = IT energy × WUE; indirect = facility energy × EWIF_eff
3. EWIF_eff = RE share × 0.1 + (1 − RE share) × grid EWIF
4. Source WUE = total water ÷ IT energy
5. Average day (kL) = IT MW × utilisation × 24 × WUE; peak day = average day × peak-day factor; peak hour (kL/h) = IT MW × utilisation × WUE × peak-hour factor
6. People's water equivalent = total litres ÷ (135 L per person per day × 365)
7. Storage (kL) = peak hour × hours of autonomy
8. Contract demand (MVA) = IT MW × hot-hour PUE ÷ 0.95
9. Power cost (₹ crore/yr) = facility MWh × 1,000 × ₹/kWh ÷ 10^7

WHAT EACH EQUATION DOES, AND WHY IT TAKES THIS FORM

- Direct water is on-site use (IT energy × WUE). Indirect water is what power plants evaporate for the electricity you buy: facility energy × effective grid intensity, where renewables count at 0.1 L/kWh.
- Peak day and peak hour are averages × peak factors for the cooling design: pipes, connections and storage must cope with peaks, not averages.
- Contract demand uses the hot-hour PUE, because the grid connection must carry the worst hour.

ASSUMED VALUES AND THE RATIONALE FOR EACH

VALUE	USED	WHY THIS VALUE (SOURCE)	BASIS
Design defaults	WUE / PUE / hot-hour PUE / peak-day / peak-hour factors per design	Report 12 calculator defaults (e.g. adiabatic 0.45 / 1.30 / 1.45 / 2.1 / 3.7). (Report 12)	Estimate
RE water	0.1 L/kWh	Panel washing for solar; wind ~0. (NREL; Report 12)	Estimate

FURTHER NOTES FROM THE ATLAS

- Design defaults (WUE, PUE, hot-hour PUE, peak factors): towers 2.0 / 1.50 / 1.65 / 1.5 / 1.7; adiabatic 0.45 / 1.30 / 1.45 / 2.1 / 3.7; dry + liquid 0.03 / 1.25 / 1.35 / 1.2 / 2.5; air-cooled chillers 0.03 / 1.60 / 1.80 / 1.1 / 2.5; seawater 0.0 / 1.25 / 1.35 / 1.0 / 1.0 [Report 12 model and proxy].
- Peak factors are relative to the annual average at the stated utilisation.
- RE-RTC (solar + wind + battery) water ≈ 0.1 L/kWh [proxy].
- Landed cost is an input; Report 12 examples range ₹6.3–11.2/kWh.

WORKED CALCULATION WITH THE SLIDER VALUES ABOVE

STEP	CALCULATION	RESULT
IT energy	50 × 80% × 8,760	**3,50,400 MWh**
Facility energy	3,50,400 × 1.3	**4,55,520 MWh**

Direct water	3,50,400 × 0.45 ÷ 1,000	**158 ML**
Effective grid water	0% × 0.1 + 100% × 3	**3.00 L/kWh**
Indirect water	4,55,520 × 3.00 ÷ 1,000	**1,367 ML**
Source WUE	(158 + 1,367) × 1,000 ÷ 3,50,400	**4.35 L/kWh**
Peak hour	50 × 80% × 0.45 × 3.7	**66.6 kL/h**
Storage	66.6 × 48 h	**3,197 kL**
Contract demand	50 × 1.45 ÷ 0.95	**76 MVA**

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28. Two buildings

On the atlas panel: *Two buildings, side by side. A 20 MW conventional colocation hall and a 100 MW AI training campus, run in India. Switch between whole-site and per-megawatt views, wet and dry cooling, and change how clean the power is.*

INPUTS TAKEN FROM THE SLIDERS AND SETTINGS

SETTING	VALUE USED
Grid water intensity	`2.6 L/kWh`
Clean power share	`0%`
Normalise	`Whole site`
Cooling	`Wet: cooling towers`

FORMULAS

1. `IT energy (MWh) = IT MW × load factor × 8,760 · colo 20 MW × 60% · AI 100 MW × 75%`
2. `Facility energy = IT energy × PUE · wet: colo 1.50, AI 1.20 · dry: colo 1.65, AI 1.30`
3. `On-site water (ML) = IT energy (GWh) × WUE · wet: colo 1.4, AI 1.0 L/kWh · dry: colo 0.1, AI 0.05 L/kWh`
4. `Grid water (ML) = facility energy (GWh) × (1 − clean share) × grid water intensity`
5. `CO₂ (kt) = facility energy (GWh) × 0.71 × (1 − clean share)`
6. `Per-MW view divides every figure by IT capacity (20 or 100 MW)`

WHAT EACH EQUATION DOES, AND WHY IT TAKES THIS FORM

- To compare fairly, both buildings use the same physics: energy = IT capacity × load × hours × PUE; on-site water = IT energy × WUE; grid water and CO_2 scale with the facility energy that is not clean.
- The AI campus runs fuller (75% vs 60%) but at a lower PUE, which is why per MW the two come out nearly equal.
- Dry cooling trades water for energy: WUE drops to ~0.05–0.1 but PUE rises by ~0.1–0.15.

ASSUMED VALUES AND THE RATIONALE FOR EACH

VALUE	USED	WHY THIS VALUE (SOURCE)	BASIS
Archetypes	`colo 20 MW at 60%; AI 100 MW at 75%`	Typical facility sizes; AI load follows LBNL (80% of hours at ~70% power). (Report 13; LBNL)	Estimate
Wet / dry PUE and WUE	`colo 1.50/1.4 → 1.65/0.1; AI 1.20/1.0 → 1.30/0.05`	Representative designs for tower vs closed-loop cooling. (Report 13)	Estimate
Grid water	`2.6 L/kWh at the plant`	MoEFCC caps 3.0–3.5 m³/MWh × ~75% coal share. (Report 13)	Estimate

FURTHER NOTES FROM THE ATLAS

- All values are Report 13's modelled archetypes [model], not measurements of a real site.
- AI load factor follows LBNL's training assumption (80% of hours at ~70% of rated power).
- Grid water 2.6 L/kWh is consumption at the power plant, from the MoEFCC 3.0 / 3.5 m³/MWh caps and a ~75% coal share; it excludes hydro reservoir evaporation. Chapter 12 uses ~3.0 L/kWh at the meter, which includes ~17% network losses; the two agree.
- Clean power (solar, wind) is counted at zero operating water and zero CO_2. Grid factor 0.71 tCO_2/MWh (CEA v21, FY25).

WORKED CALCULATION WITH THE SLIDER VALUES ABOVE

STEP	CALCULATION	RESULT
Colo IT energy	`20 × 60% × 8.76`	**105.1 GWh**
Colo facility energy	`105.1 × 1.5`	**157.7 GWh**
Colo on-site water	`105.1 × 1.4`	**147 ML**
Colo grid water	`157.7 × 1.00 × 2.6`	**410 ML**
Colo CO_2	`157.7 × 0.71 × 1.00`	**112 kt**
AI IT energy	`100 × 75% × 8.76`	**657.0 GWh**
AI facility energy	`657.0 × 1.2`	**788.4 GWh**
AI on-site water	`657.0 × 1`	**657 ML**
AI grid water	`788.4 × 1.00 × 2.6`	**2,050 ML**
AI CO_2	`788.4 × 0.71 × 1.00`	**560 kt**

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29. Rack density

On the atlas panel: *How hot is one rack?. kW per rack on a log scale. Solid bars are reported, pale bars estimates, hatched bars design targets.*

INPUTS

No sliders: this calculation uses fixed model data.

FORMULAS

1. `Bars span the reported low–high kW per rack; log scale 1–1,000 kW`
2. `Equivalent racks = AI rack kW ÷ conventional rack kW (rounded)`

WHAT EACH EQUATION DOES, AND WHY IT TAKES THIS FORM

- Racks of equal power make equal heat; dividing AI rack kW by conventional rack kW shows how much heat is packed into one cabinet, which is why air stops working and liquid takes over.

ASSUMED VALUES AND THE RATIONALE FOR EACH

VALUE	USED	WHY THIS VALUE (SOURCE)	BASIS
Conventional rack	`8 kW default`	Typical Indian colo rack today. (Report 13)	Estimate

FURTHER NOTES FROM THE ATLAS

- GB200 NVL72: 120 kW (NVIDIA reference) to 132 kW (HPE: 115 kW liquid + 17 kW air). GB300 up to 142 kW. Fairwater built for ~140 kW per rack.
- Vera Rubin VR200 ~190–230 kW is a third-party estimate; NVIDIA has not disclosed it. Kyber 600 kW is a vendor roadmap statement for 2027.

WORKED CALCULATION WITH THE SLIDER VALUES ABOVE

STEP	CALCULATION	RESULT
Equivalent racks	`120 ÷ 8`	**15**

13 · AI VS CONVENTIONAL

30. Per prompt

On the atlas panel: *One AI prompt: energy and water. The figures disagree by up to ~170×, mostly because each counts different things. The coloured chips show what each number includes.*

INPUTS TAKEN FROM THE SLIDERS AND SETTINGS

SETTING	VALUE USED
Measure	Energy per prompt

FORMULAS

1. `LED time (s) = Wh × 3,600 ÷ 10 W`
2. `Drops = mL ÷ 0.05 mL per drop`
3. `Gemini + India grid water = 0.26 mL + 0.24 Wh × 2.6 L/kWh = 0.26 + 0.62 ≈ 0.88 mL`
4. `Log scales: energy 0.05–100 Wh; water 0.05–100 mL`

WHAT EACH EQUATION DOES, AND WHY IT TAKES THIS FORM

- Per-prompt numbers differ mostly by what they count: chip only, the whole server and idle machines, on-site cooling water, power-plant water or chip manufacturing. Putting them on one log scale shows how far apart the boundaries put them.
- Indian grid water is added because Google's 0.26 mL counts only on-site water; in India the power plants add ~2.6 L per kWh.

ASSUMED VALUES AND THE RATIONALE FOR EACH

VALUE	USED	WHY THIS VALUE (SOURCE)	BASIS
Drop	`0.05 mL`	Standard drop volume. (Convention)	Convention
LED	`10 W`	Household LED bulb. (Convention)	Convention

FURTHER NOTES FROM THE ATLAS

- Google: median Gemini Apps text prompt, May 2025, full stack including idle capacity and PUE 1.09; chip-only method gives 0.10 Wh / 0.12 mL.
- OpenAI 0.34 Wh and ~0.32 mL are an unaudited CEO statement (June 2025) with undisclosed scope. Epoch AI values are modelled for GPT-4o on H100.
- Li et al. "10–50 mL" = one 500 mL bottle per 10–50 responses, on-site + grid water, GPT-3-era hardware. Mistral 45 mL is a full life-cycle figure (ADEME / Carbone 4 method).
- None of the disclosed medians cover reasoning models, long contexts, image or video generation.

WORKED CALCULATION WITH THE SLIDER VALUES ABOVE

STEP	CALCULATION	RESULT
Gemini + Indian grid water	`0.26 mL + 0.24 Wh × 2.6 L/kWh`	**`0.26 + 0.62 = 0.88 mL`**
LED time for 0.24 Wh	`0.24 × 3,600 ÷ 10 W`	**`86 s`**
Drops in 0.26 mL	`0.26 ÷ 0.05`	**`≈ 5 drops`**

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31. Going dry

On the atlas panel: *Does liquid cooling save water? Run the trade. Liquid cooling lets heat leave through dry coolers instead of evaporating it. That costs extra electricity, and in India extra electricity carries coal-plant water and CO_2.*

INPUTS TAKEN FROM THE SLIDERS AND SETTINGS

SETTING	VALUE USED
Wet design WUE	`1.00 L/kWh`

PUE penalty for going dry	+0.10
Grid water intensity	2.6 L/kWh
Clean power share	0%
Site	100 MW AI campus

FORMULAS

1. IT energy (GWh) = MW × load factor × 8.76
2. On-site water saved (ML) = IT GWh × (WUE wet − WUE dry)
3. Extra electricity (GWh) = IT GWh × PUE penalty
4. Extra grid water (ML) = extra GWh × (1 − clean share) × grid L/kWh · extra CO_2 (kt) = extra GWh × 0.71 × (1 − clean share)
5. Net water = saved − extra grid water
6. Break-even grid intensity (L/kWh) = (WUE wet − WUE dry) ÷ PUE penalty; going dry saves water whenever the grid is below this

WHAT EACH EQUATION DOES, AND WHY IT TAKES THIS FORM

- Going dry saves the water the towers would have evaporated (IT energy × the WUE difference) but adds electricity (IT energy × the PUE penalty).
- In India that extra electricity has its own water (grid L/kWh) and CO_2 (0.71 t/MWh) unless it is clean.
- Break-even: going dry saves water whenever the grid's water per kWh is below the WUE saved ÷ PUE penalty.

ASSUMED VALUES AND THE RATIONALE FOR EACH

VALUE	USED	WHY THIS VALUE (SOURCE)	BASIS
Dry WUE	0.05 L/kWh	Closed loop with small top-up. (Report 13)	Estimate
PUE penalty default	+0.10	Typical energy cost of removing evaporation. (Report 13; Microsoft "nominal increase")	Estimate

FURTHER NOTES FROM THE ATLAS

- Defaults reproduce Report 13 section 4: 100 MW AI at 75% load, WUE 1.0 → 0.05 L/kWh, PUE penalty 0.10, grid 2.6 L/kWh: −624 ML on site, +171 ML grid, net −453 ML, +47 kt CO_2.
- Dry WUE fixed at 0.05 L/kWh (closed loop, small top-up).

WORKED CALCULATION WITH THE SLIDER VALUES ABOVE

STEP	CALCULATION	RESULT
IT energy	100 × 75% × 8.76	**657.0 GWh**
On-site saved	657.0 × (1.00 − 0.05)	**624 ML**
Extra electricity	657.0 × 0.10	**65.7 GWh**
Extra grid water	65.7 × 1.00 × 2.6	**171 ML**
Net water	171 − 624	**-453 ML**
Extra CO_2	65.7 × 0.71 × 1.00	**46.6 kt**
Break-even grid water	(1.00 − 0.05) ÷ 0.10	**9.5 L/kWh**

14 · Least-water power

14 · LEAST-WATER POWER

32. Hourly linear programme

On the atlas panel: *Cost against water: every option the model found. Each dot is a least-cost mix for a given water cap. Moving left saves water; moving up costs more. Points below the dashed line save water for less than the city charges*

INPUTS TAKEN FROM THE SLIDERS AND SETTINGS

SETTING	VALUE USED
Value of water	₹99/kL
Hydro water case	10 L/kWh · base

FORMULAS

1. Decision variables: solar MW (Cs), wind MW (Cw), hydro contract MW (Ch ≤ 40), battery MWh (E); and for each of 8,760 hours hydro h, grid g, charge c, discharge d, state of charge q, curtailment u
2. Energy balance, every hour: Cs·S(t) + Cw·W(t) + h + d + g − c − u = 65 MW
3. Battery: q(t+1) = q(t) + 0.938·c − d ÷ 0.938 (88% round trip, cyclic); q ≤ E; c, d ≤ E ÷ 4 (4-hour unit)
4. Hydro: h ≤ Ch each hour; monthly energy ≤ Ch × monthly capacity factor × hours (45% Jan … 85% Aug)
5. Cost (₹ crore/yr) = Cs × (3.8 × CRF(10%, 25) + 1.5% O&M) + Cw × (7.0 × CRF + 1.5%) + Ch × ₹4.5/kWh take-or-pay + E × ₹0.105 cr/MWh + ₹1.0/kWh open-access charge on delivered RE + ₹7.0/kWh on grid energy
6. Water (ML/yr) = 0.1 L/kWh × solar output + hydro L/kWh × hydro energy + 2.0 L/kWh × grid energy
7. Frontier: minimise cost; minimise water; then minimise cost with water ≤ min + f × (cheapest − min), f = 0.75 … 0.05
8. ₹ per kL saved = (cost − cheapest cost) ÷ (cheapest water − water)
9. "Water cap N%": water allowed = minimum water + N% × (cheapest-mix water − minimum water)

WHAT EACH EQUATION DOES, AND WHY IT TAKES THIS FORM

- The model is a linear programme: it chooses capacities and an hourly dispatch to minimise cost, subject to meeting the 65 MW load every hour and the battery's physics. Linear programming finds the true optimum, not a guess.
- To see the trade-off, it first finds the cheapest mix and the lowest-water mix, then re-solves for the cheapest mix under a series of water caps between them. Each dot is one of those solutions.
- Cost and water are sums of simple parts, shown below for the recommended mix: annualised capacity costs, ₹1/kWh open access on renewables used, ₹7/kWh on grid energy; and 0.1 L/kWh for solar washing, 10 L/kWh for reservoir hydro, 2.0 L/kWh for grid power.

ASSUMED VALUES AND THE RATIONALE FOR EACH

VALUE	USED	WHY THIS VALUE (SOURCE)	BASIS
Load	50 MW IT × PUE 1.30 = 65 MW flat	An efficient mid-size facility. (Model)	Series assumption
Solar / wind CF	20.5% / 33%	Pavagada-like solar; Chitradurga/Gadag-like wind (synthetic profiles). (Model)	Estimate
Capex	solar ₹3.8 cr/MW, wind ₹7.0 cr/MW, 25 yrs, 10%, O&M 1.5%	2026 Indian benchmarks. (Model)	Estimate
Battery	₹3.5 lakh per MW-month (4-hour unit), 88% round trip	Recent Indian BESS tender prices. (Model)	Estimate

Prices	`hydro ₹4.5/kWh take-or-pay; grid ₹7.0/kWh; open access ₹1.0/kWh`	Indicative Karnataka contract and tariff levels. (Model)	Estimate
Water	`solar 0.1, grid 2.0, hydro 0/10/20 L/kWh`	Panel washing; coal cooling; reservoir evaporation (unmeasured, hence three cases). (Model)	Estimate

FURTHER NOTES FROM THE ATLAS

- Solar profile: Pavagada/Tumakuru-like, capacity factor 20.5%, monsoon dip. Wind: Chitradurga/Gadag-like, CF 33%, peaking June–August with an evening bias. Both synthetic (seeded random weather), not measured data.
- Hydro water: base 10 L/kWh (reservoir evaporation), run-of-river 0, high 20. Karnataka grid 2.0 L/kWh consumptive. Solar 0.1 L/kWh (panel washing).
- Flat 65 MW load (50 MW IT × PUE 1.30). CO_2 = grid MWh × 0.71 t (CEA); renewables counted at zero.
- Solved with SciPy HiGHS; each solve ~35 s, 52,564 variables.

WORKED CALCULATION WITH THE SLIDER VALUES ABOVE

STEP	CALCULATION	RESULT
Solar	`161 MW × ₹0.480 cr`	**₹77.4 cr**
Wind	`86 MW × ₹0.880 cr`	**₹75.7 cr**
Battery	`379 MWh × ₹0.1 cr`	**₹37.9 cr**
Open access	`(282 + 249 − 49 curtailed) GWh × ₹1`	**₹48.2 cr**
Grid energy	`94.3 GWh × ₹7`	**₹66.0 cr**
Total	`sum (model: ₹306.7 cr; rounding < 1%)`	**₹5.39/kWh over 569 GWh**
Water	`282 GWh × 0.1 + 94.3 GWh × 2.0`	**218 ML**
₹ per kL saved	`(306.7 − 296.2) × 10⁷ ÷ ((1,451 − 218) × 1,000)`	**₹85**

14 · LEAST-WATER POWER

33. Price of water saved

On the atlas panel: *Rule 1 · Save water only while it is cheaper than buying it. Extra cost of each option per kilolitre of water saved, compared with the cheapest mix (base case).*

INPUTS

No sliders: this calculation uses fixed model data.

FORMULAS

1. `Average: (cost − cheapest cost) × 10⁷ ÷ ((cheapest water − water) × 1,000) in ₹/kL`
2. `Step: the same between neighbouring points on the frontier`

WHAT EACH EQUATION DOES, AND WHY IT TAKES THIS FORM

- The value of saving water is compared with what the water would cost to buy. If a kilolitre saved costs less than tap water, saving it is worth it on cost alone.
- The comparison is against the cheapest mix, so each option's extra cost is set against the water it saves relative to that mix.

ASSUMED VALUES AND THE RATIONALE FOR EACH

VALUE	USED	WHY THIS VALUE (SOURCE)	BASIS
Water value	₹99/kL	BWSSB rate used in the simulator; tanker water costs more, so the test is conservative. (BWSSB)	Estimate

FURTHER NOTES FROM THE ATLAS

- ₹99/kL municipal benchmark is the BWSSB rate used in the atlas simulator; tanker water costs more, so the rule is conservative.
- Carbon co-benefits are not priced in this rule; they appear in Rule 3.

WORKED CALCULATION WITH THE SLIDER VALUES ABOVE

STEP	CALCULATION	RESULT
Water cap 75%	(₹296.5 − ₹296.2 cr) × 10^7 ÷ ((1,451 − 1,088) ML × 1,000)	**₹8/kL ≤ ₹99** ✓
Water cap 50%	(₹297.6 − ₹296.2 cr) × 10^7 ÷ ((1,451 − 725) ML × 1,000)	**₹19/kL ≤ ₹99** ✓
Water cap 35%	(₹298.8 − ₹296.2 cr) × 10^7 ÷ ((1,451 − 508) ML × 1,000)	**₹27/kL ≤ ₹99** ✓
Water-only knee	(₹299.7 − ₹296.2 cr) × 10^7 ÷ ((1,451 − 363) ML × 1,000)	**₹32/kL ≤ ₹99** ✓
Recommended	(₹306.7 − ₹296.2 cr) × 10^7 ÷ ((1,451 − 218) ML × 1,000)	**₹85/kL ≤ ₹99** ✓
Water cap 10%	(₹314.9 − ₹296.2 cr) × 10^7 ÷ ((1,451 − 145) ML × 1,000)	**₹143/kL > ₹99** ✗
Near-zero water	(₹330.6 − ₹296.2 cr) × 10^7 ÷ ((1,451 − 73) ML × 1,000)	**₹249/kL > ₹99** ✗

15 · Recommendations

15 · RECOMMENDATIONS

34. Hub ranking

On the atlas panel: *Which hub, for which kind of data centre?. Eleven hubs scored 1–5 on six criteria (5 = best). Pick an archetype or set your own weights; the ranking re-sorts. Tap a hub for the design response it needs.*

INPUTS TAKEN FROM THE SLIDERS AND SETTINGS

SETTING	VALUE USED
Power headroom	25
Clean power	10
Water	15
Hazard	10
Latency / cables	20
Land	10
Cloud: strict latency rule (cable-landing metros, score ≥ 4)	on
Archetype	Cloud / colocation

FORMULAS

1. Score (out of 5) = Σ (criterion score × weight) ÷ Σ weights
2. Cloud/colo weights: power 25, clean power 10, water 15, hazard 10, latency 20, land 10
3. AI training weights: power 30, clean power 20, water 15, hazard 10, latency 5, land 10
4. Latency rule (cloud only, optional): drop hubs scoring below 4 on latency/cable landing

WHAT EACH EQUATION DOES, AND WHY IT TAKES THIS FORM

- A weighted average combines six 1–5 scores into one: multiply each by its weight, add, divide by the sum of weights. The result stays on the 1–5 scale.
- Cloud weights latency heavily (20%) because users notice delay; AI training weights power (30%) and clean power (20%) because it is latency-tolerant but power-hungry.

ASSUMED VALUES AND THE RATIONALE FOR EACH

VALUE	USED	WHY THIS VALUE (SOURCE)	BASIS
Hub scores	1–5 on six criteria	Report 15 estimates; tier-2 is a group average. (Report 15)	Estimate
Weights	cloud 25/10/15/10/20/10; AI 30/20/15/10/5/10	Report 15 framework (policy and skills not scored per hub). (Report 15)	Series assumption

FURTHER NOTES FROM THE ATLAS

- Hub scores are Report 15 estimates, not measured indices; tier-2 is a group average.
- The full framework also weights policy (5%) and skills (5%); they are not scored per hub, so weights are re-normalised over the six criteria.
- Use the chapter 10 ranking engine for city-level detail.

WORKED CALCULATION WITH THE SLIDER VALUES ABOVE

STEP	CALCULATION	RESULT
Hub	Hyderabad	
Power headroom	4 × 25	**100.0**

Clean power	3 × 10	**30.0**
Water	3 × 15	**45.0**
Hazard	5 × 10	**50.0**
Latency / cables	3 × 20	**60.0**
Land	4 × 10	**40.0**
Score	325.0 ÷ 90	**3.61 / 5**

15 · RECOMMENDATIONS

35. Design check

On the atlas panel: *Check a design against the targets. Pick an archetype and climate, then enter your design. Each bar shows how far you are from the recommended target, with international rules marked for comparison.*

INPUTS TAKEN FROM THE SLIDERS AND SETTINGS

SETTING	VALUE USED
Your annual PUE	1.30
Your WUE (L/kWh)	0.60 L/kWh
Hourly clean power	40%
Archetype	AI training campus
Climate	Mild

FORMULAS

1. Pass if PUE ≤ target(archetype, climate), WUE ≤ target, hourly CFE ≥ target
2. CUE (kg CO_2 per IT-kWh) = PUE × 0.71 × (1 − CFE), counting clean power at zero
3. Benchmarks: IndiaAI tender PUE < 1.35; Germany EnEfG new builds ≤ 1.2 (from July 2026); Singapore DC-CFA2 ≤ 1.25; China new large DCs ≤ 1.25

WHAT EACH EQUATION DOES, AND WHY IT TAKES THIS FORM

- Targets differ by climate because hot, humid air makes cooling harder: the same design reaches a lower PUE in Bengaluru than in Chennai.
- CUE (carbon per kWh of IT) = PUE × grid factor × the share not covered by clean power, so it rewards efficiency and clean power together.

ASSUMED VALUES AND THE RATIONALE FOR EACH

VALUE	USED	WHY THIS VALUE (SOURCE)	BASIS
Targets	by archetype and climate	Report 15 engineering estimates benchmarked to Germany (1.2), Singapore (1.25), China (1.25) and IndiaAI (1.35). (Report 15)	Estimate

FURTHER NOTES FROM THE ATLAS

- Targets are Report 15 engineering estimates by climate: mild (Bengaluru, Pune, Hyderabad), hot-humid (Mumbai, Chennai, Kolkata, Vizag), hot-arid (NCR, Jamnagar, Rajasthan, GIFT City).
- WUE is site consumption per kWh of IT energy; CFE is the share of hours matched with clean power by 2030.

WORKED CALCULATION WITH THE SLIDER VALUES ABOVE

STEP	CALCULATION	RESULT
Target lookup	AI training campus, mild climate	**PUE ≤ 1.15 · WUE ≤ 0.2 · CFE ≥ 80% · CUE ≤ 0.15**
PUE	1.30 vs ≤ 1.15	**fail ✗**

WUE	0.60 vs ≤ 0.2	**fail ✗**
Hourly clean power	40% vs ≥ 80%	**fail ✗**
CUE	1.30 × 0.71 × (1 − 40%) = 0.55 vs ≤ 0.15	**fail ✗**

15 · RECOMMENDATIONS

36. How to read the policy-tools grid

On the atlas panel: *The tools other countries use, and India's gaps. Peers have converged on six tools. Tap a cell for the rule behind it.*

INPUTS

No sliders: this calculation uses fixed model data.

FORMULAS

1. ● in force · ◐ partial, draft or indirect · — none
2. India row: the best national instrument found today

WHAT EACH EQUATION DOES, AND WHY IT TAKES THIS FORM

- Each cell codes whether a jurisdiction has the tool in force (●), partly or in draft (◐), or not at all (—). India's row shows the nearest national instrument, which makes the gaps visible.

ASSUMED VALUES AND THE RATIONALE FOR EACH

VALUE	USED	WHY THIS VALUE (SOURCE)	BASIS
Coding	● / ◐ / –	In force / partial or draft / none, from the research report. (Report 15)	Series assumption

FURTHER NOTES FROM THE ATLAS

- Netherlands and Virginia entries were not re-verified in this research; Johor WUE thresholds are reported inconsistently (1.4 vs 1.8).
- China: NDRC 2024 plan, ≤1.25 for new large DCs, ≤1.2 at hub nodes.

WORKED RESULT

Rule-based: the result is the branch of the rules above that matches the inputs; there is no arithmetic.